QUANT SERIES · PART 5 OF 8

Stochastic Calculus, Step by Step

Part 5: the maths toolkit, from Cholesky to ADI



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 5 OF 8

Stochastic Calculus, Step by Step

Part 5: The maths toolkit, with Python

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Stochastic Calculus, Step by Step. Part 5: The maths toolkit, with Python.

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One set of numbers runs through the whole course: a share priced at $S_0 = 100$, a strike $K = 105$, a risk-free rate $r = 3\%$ (continuously compounded), a volatility $\sigma = 20\%$ and a horizon $T = 1$ year. Part 5 adds Lesson 27's second share ($S_0 = 95$, $\sigma = 30\%$, correlation 0.5) and Lesson 28's Vasicek rate ($\kappa = 0.5$, $\bar{r} = 4\%$, $s = 1\%$, $r_0 = 3\%$), and a few small made-up curves and quotes. They are all made up for teaching, and every random path is simulated with a fixed seed. There is no market data.

The exercises are at the end of each lesson. Their full solutions, and the answers to the self-test questions, are in the companion volume, Stochastic Calculus, Step by Step, Part 5: Solutions.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only), one per lesson, which reproduce each result. Every algorithm is written out in full rather than called from a library.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This is Part 5 of *Stochastic Calculus, Step by Step*, a course that goes from school calculus to the Black-Scholes formula and on to the models banks use for currencies and interest rates. Parts 1 to 4 built the calculus of one random number at a time and used it to price options. The parts that follow need more: several correlated shares, interest rates and volatilities that move, prices computed from characteristic functions, smiles fitted to quotes, regressions inside simulations, and equations in two dimensions. **Part 5 builds those tools**, once and by hand, on problems you have already met.

LESSON	TOPIC	WHAT YOU TAKE AWAY
31	Linear algebra for several risks	Covariance matrices, which correlation tables are possible, Cholesky, eigenvalues and PCA
32	Ordinary differential equations	The integrating factor, Riccati equations, Runge-Kutta, and solutions that blow up
33	Complex numbers and Fourier transforms	Characteristic functions, and a price computed by Fourier inversion
34	Numerical methods	Implied volatility by Newton and Brent, splines, Gaussian quadrature, Levenberg-Marquardt
35	Regression and PDEs, further	Regression as conditional expectation, kernels, the forward equation, ADI in two dimensions

How Part 5 checks itself

Every lesson ends by getting a number you already know by a new route. Lesson 31's hand-written Cholesky reprices Lesson 30's five-share basket to the last digit. Lesson 32's Riccati equation gives a bond price that a simulation confirms. Lesson 33 prices the course's call, 7.1281, without ever writing down a density. Lesson 34 recovers the 20% volatility from that price. Lesson 35 finds Margrabe's exchange-option price from Lesson 27 on a two-dimensional grid. When a new tool agrees with an old answer, you can trust it on a problem where there is no old answer.

Who this book is for

As in the earlier parts, this book is written for readers learning on their own, without a teacher to fill the gaps. Every new idea is **defined** before it is used, **explained** in words, tried on a small example, and then **derived** one line at a time, with the reason for each line. When a step uses an earlier lesson, the lesson is named.

You need Parts 1 to 4. In particular: covariance and the normal distribution (Lessons 3 and 4), Taylor's theorem (Lesson 1), the Vasicek bond (Lesson 28), the Black-Scholes formula and vega (Lessons 20 and 21), Longstaff-Schwartz (Lesson 30), finite differences and the tridiagonal solver (Lesson 29), and Margrabe's formula (Lesson 27). No linear algebra, complex numbers or numerical analysis is assumed: each is built from the start.

One set of numbers

The course's numbers are unchanged: a share at $S_0 = 100$, a strike $K = 105$, a volatility $\sigma = 20\%$, a risk-free rate $r = 3\%$ (continuously compounded) and a horizon of $T = 1$ year, with the call worth 7.1281. Part 5 also uses Lesson 27's second share ($S_0 = 95$, $\sigma = 30\%$, correlation 0.5 with the first) and Lesson 28's Vasicek rate (speed $\kappa = 0.5$, long-run level $\bar{r} = 4\%$, volatility $s = 1\%$, today's rate $r_0 = 3\%$). A few small curves and quotes are made up where a lesson needs them. Every random path is simulated with a fixed seed.

How each lesson is built

Every lesson follows the same steps as before: **why this lesson**, **what you need** from earlier, **the picture**, the derivations as numbered steps, **numbers and code**, **what to remember**, **common mistakes**, eight **exercises** (six by hand, two in code) and a three-question **self-test**.

How to read it

Part 5 is a toolkit, and toolkits are best learned by using the tools. The notebooks write out every algorithm in full, in a few lines of numpy: Cholesky, Runge-Kutta, Brent's method, the cubic spline, Levenberg-Marquardt, the kernel regression and the ADI scheme. Run them, change a number, and see whether the result moves the way the text says. Several of the results are cautionary, and worth seeing for yourself: Newton's

method, the fastest root finder, fails badly on a deep out-of-the-money option; a cubic spline through five sensible zero rates draws a hump in the forward curve; and the simplest ADI scheme loses an order of accuracy as soon as two shares are correlated.

Notebooks and solutions

Every number, table and figure is computed by a companion Python notebook, one per lesson, using numpy only. The notebooks are free at github.com/RatesWithBip/curriculum.

The solutions are **not** in this book: they are in the companion volume, *Part 5: Solutions*, with every derivation written out and every code solution shown with what it prints. Try each exercise before you look.

Reading alongside

Gilbert Strang's *Introduction to Linear Algebra* covers Lesson 31 in far more depth; *Numerical Recipes* by Press, Teukolsky, Vetterling and Flannery covers Lessons 32 and 34; Jim Gatheral's *The Volatility Surface* uses Lesson 33's Fourier pricing for the Heston model; and Daniel Duffy's *Finite Difference Methods in Financial Engineering* goes further with Lesson 35's ADI schemes. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and both books are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

LESSON 31 · VIDEO 31

Linear Algebra for Several Risks

Portfolios as vectors and the covariance matrix; why a correlation table must be positive semi-definite, and an innocent-looking one that isn't; Cholesky, which turns independent normals into correlated ones; eigenvalues, principal components and how to repair a broken table; and Lesson 30's five-share basket, repriced with a hand-written Cholesky.

Why this lesson

One share needs one random number per step. Five shares need five, and they must move together the way the shares do: a market fall pulls most of them down at once. So two questions come before any multi-asset model:

- **How do we describe how several risks move together?** With a covariance matrix: one table that holds every variance and every covariance.
- **How do we simulate them?** By turning independent normals, which are easy to draw, into correlated ones with the right covariance matrix.

Both answers are linear algebra. This lesson builds the little of it that finance uses again and again: matrix products, positive definiteness, the **Cholesky factor** and **eigenvalues**. It is used in Lessons 36 to 38 (several shares), 49 (implied correlation) and 66 (two-factor interest-rate models).

What you need from earlier lessons

- **Variance and covariance** (Lesson 3). $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$, and $\mathrm{Cov}(X, Y) = \rho\,\mathrm{sd}(X)\,\mathrm{sd}(Y)$.
- **Sums of normals** (Lesson 4). Independent normals add up to a normal.
- **Two correlated Brownian motions** (Lesson 14, Exercise 14.6). $B = \rho W + \sqrt{1-\rho^2}\,W'$, with W' independent of W, has correlation ρ with W.
- **The basket call** (Lesson 30). A call on the average of five shares, each with $\sigma = 20\%$ and pairwise correlation 0.5: 5.3391.

Numbers. Two shares: the course's (σ_1 = 20%) and Lesson 27's second share (σ_2 = 30%), with correlation $\rho = 0.5$.

The picture

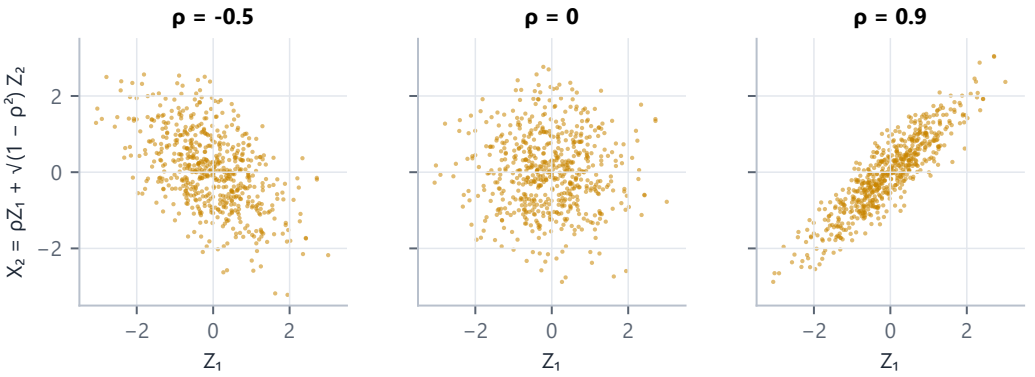


Figure 31.1. 600 pairs (Z_1, X_2), where $X_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2$ is built from two independent standard normals. The same Z's give a falling cloud, a round one and a tight rising one, depending only on ρ.

Every correlated simulation in finance is built this way: draw independent normals, then mix them. The recipe for the mixing, with any number of shares, is the Cholesky factor.

Vectors, matrices and the portfolio's variance

Vectors and matrices. A **vector** is a list of numbers, written as a column: the weights of a portfolio, $w = (w_1, w_2)^\top$, or the returns of its shares, $R = (R_1, R_2)^\top$. The **transpose** $^\top$ turns a column into a row. A **matrix** is a table of numbers. Multiplying a row by a column adds up the products of matching entries: $w^\top R = w_1 R_1 + w_2 R_2$, the portfolio's return. Multiplying a matrix by a vector does this for each row in turn.

The covariance matrix. For returns $R_1, \ldots, R_n$, the **covariance matrix** Σ has $\Sigma_{ij} = \mathrm{Cov}(R_i, R_j)$. Its diagonal holds the variances ($\mathrm{Cov}(R_i, R_i) = \mathrm{Var}(R_i)$), and it is symmetric, $\Sigma_{ij} = \Sigma_{ji}$. For our two shares:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} 0.04 & 0.03 \\ 0.03 & 0.09 \end{pmatrix}.$$

Claim. The portfolio's variance is $\mathrm{Var}(w^\top R) = w^\top \Sigma w$.

Step 1. Expand. $w^\top R = \sum_i w_i R_i$, and covariance is linear in each argument (Lesson 3), so

$$\operatorname{Var}\Big(\sum_i w_i R_i\Big) = \operatorname{Cov}\Big(\sum_i w_i R_i, \sum_j w_j R_j\Big) = \sum_i \sum_j w_i w_j \operatorname{Cov}(R_i, R_j).$$

Step 2. Recognise the product. $\sum_j w_j \Sigma_{ij}$ is entry i of Σw, and multiplying by w_i and adding over i is $w^\top(\Sigma w)$. ■

In numbers, half in each share, $w = (0.5, 0.5)$:

$$\begin{aligned} w^\top \Sigma w &= 0.25 \times 0.04 + 0.25 \times 0.09 + 2 \times 0.25 \times 0.03 \\ &= 0.0100 + 0.0225 + 0.0150 = 0.0475, \end{aligned}$$

a volatility of 0.2179. In 200,000 simulated pairs the variance is 0.0474. Notice that 21.8% is less than the average of the two volatilities, 0.25: the shares don't always move together, so some of their moves cancel. That is **diversification**, and the off-diagonal entries of Σ measure how much of it there is. The least risky mix puts 0.857 of the money in the first share, for a volatility of 0.1964: lower than either share on its own (Exercise 31.1).

Which correlation tables are possible

Definition. A symmetric matrix A is **positive semi-definite** if $w^\top A w \geq 0$ for every vector w, and **positive definite** if $w^\top A w > 0$ for every $w \neq 0$.

Every covariance matrix is positive semi-definite. The reason is one line: $w^\top \Sigma w$ is the variance of the portfolio $w^\top R$, and a variance can't be negative. A correlation matrix is the covariance matrix of the standardised returns $R_i / \operatorname{sd}(R_i)$, so the same holds for it.

The converse is the trap. Fill in a table of correlations by hand, or estimate each pair from a different stretch of data, and you can get something that is no correlation matrix at all. Here is one: shares 1 and 2 move together ($\rho_{12} = 0.9$), shares 2 and 3 move together ($\rho_{23} = 0.9$), but shares 1 and 3 move in opposite directions ($\rho_{13} = -0.9$). Each entry is between −1 and 1. But try the portfolio $w = (1, -1, 1)$:

$$w^\top C w = \underbrace{1 + 1 + 1}_{\text{variances}} + 2\big(\underbrace{-0.9}_{w_1 w_2 \rho_{12}} \; \underbrace{-0.9}_{w_1 w_3 \rho_{13}} \; \underbrace{-0.9}_{w_2 w_3 \rho_{23}} \big) = 3 - 5.4 = -2.4.$$

A negative variance. The table is impossible: if 1 moves with 2 and 2 moves with 3, then 1 can't move strongly against 3. Its determinant is -2.888, and keeping $\rho_{12} = \rho_{23} = 0.9$, only ρ_{13} between 0.62 and 1 gives a valid table (Exercise 31.3).

Why it matters. A Monte Carlo built on an invalid table either crashes (the Cholesky factor below doesn't exist) or, worse, quietly uses a different table and prices something nobody asked for.

Cholesky: correlated draws from independent ones

Definition. A **Cholesky factor** of a positive definite matrix Σ is a lower-triangular matrix L (zeros above the diagonal) with positive diagonal entries and $LL^\top = \Sigma$.

Claim. If $Z = (Z_1, \dots, Z_n)^\top$ are independent standard normals, then $X = LZ$ has covariance matrix Σ.

Step 1. Each $X_i = \sum_k L_{ik} Z_k$ has mean 0, so $\operatorname{Cov}(X_i, X_j) = E[X_i X_j]$.

Step 2. $E[X_i X_j] = \sum_k \sum_l L_{ik} L_{jl} E[Z_k Z_l]$. Independence gives $E[Z_k Z_l] = 1$ if $k = l$ and 0 otherwise, so only $k = l$ survives:

$$E[X_i X_j] = \sum_k L_{ik} L_{jk} = (LL^\top)_{ij} = \Sigma_{ij}. \quad \blacksquare$$

Two shares, by hand. Write $L = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$, so that $LL^\top = \begin{pmatrix} a^2 & ab \\ ab & b^2 + c^2 \end{pmatrix}$, and match entries with Σ:

- $a^2 = 0.04$, so $a = 0.2$;
- $ab = 0.03$, so $b = 0.15$;
- $b^2 + c^2 = 0.09$, so $c = \sqrt{0.09 - 0.0225} = 0.2598$.

So $X_1 = 0.2 Z_1$ and $X_2 = 0.15 Z_1 + 0.2598 Z_2 = 0.3\big(0.5 Z_1 + \sqrt{1 - 0.5^2}\, Z_2\big)$: Lesson 14's recipe, scaled by the volatilities. The notebook's routine agrees with numpy's to 0.0e+00.

Any number of shares. Match the entries of $LL^\top$ with Σ one column at a time. Entry (j, j) involves only row j of L, and entry (i, j) below the diagonal involves rows i and j:

$$L_{jj} = \sqrt{\Sigma_{jj} - \sum_{k<j} L_{jk}^2}, \qquad L_{ij} = \frac{1}{L_{jj}} \Big(\Sigma_{ij} - \sum_{k<j} L_{ik} L_{jk} \Big) \quad (i > j).$$

Everything on the right was computed in earlier columns.

Cholesky, one column at a time.

```
def cholesky(A):
    # Lower-triangular L with L @ L.T = A, column by column. Fails (raises) if A is not posi-
tive definite.
    n = len(A)
    L = np.zeros((n, n))
    for j in range(n):
        d = A[j, j] - L[j, :j] @ L[j, :j]
        if d <= 0:
            raise ValueError(f"not positive definite: pivot {j + 1} is {d:.4f}")
        L[j, j] = math.sqrt(d)
        for i in range(j + 1, n):
            L[i, j] = (A[i, j] - L[i, :j] @ L[j, :j]) / L[j, j]
    return L
```

Three shares. For the correlation matrix with $\rho_{12} = 0.5$, $\rho_{13} = 0.3$, $\rho_{23} = 0.4$, the recipe gives (Exercise 31.5)

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0.5 & 0.8660 & 0 \\ 0.3 & 0.2887 & 0.9092 \end{pmatrix},$$

and $LL^\top$ reproduces the table to 1.1e-16. With volatilities 20%, 30% and 25%, 400,000 simulated triples have a covariance matrix within 2.0e-04 of the target.

Cholesky as a test. On the impossible table, the recipe breaks at the third column: not positive definite: pivot 3 is -15.2000. A square root of a negative number is the matrix's way of saying it isn't positive definite. Computing a Cholesky factor is the cheapest way to check.

Eigenvalues and principal components

Definition. A non-zero vector v is an **eigenvector** of a matrix A, with **eigenvalue** λ, if $Av = \lambda v$: the matrix only stretches it, without turning it.

The spectral theorem (stated, not proved here). A symmetric $n \times n$ matrix has n real eigenvalues and n eigenvectors at right angles to one another. Collecting the eigenvectors (each of length 1) as the columns of Q and the eigenvalues on the diagonal of Λ:

$$\Sigma = Q \Lambda Q^\top, \qquad Q^\top Q = I.$$

What it means for risk. Change coordinates to $y = Q^\top w$. Then $w^\top \Sigma w = y^\top \Lambda y = \sum_k \lambda_k y_k^2$. In the eigenvector directions the risks are **uncorrelated**, and each eigenvalue is the variance along its direction. These directions are the **principal components**, and listing them in order of their eigenvalues is **principal component analysis**, or PCA.

And the test for a valid table. $\sum_k \lambda_k y_k^2 \geq 0$ for every y exactly when every $\lambda_k \geq 0$. So: **a symmetric matrix is positive semi-definite if and only if all its eigenvalues are non-negative.** The impossible table has eigenvalues -0.8, 1.9 and 1.9, and its eigenvector for -0.8 is, up to scale, the portfolio $(1, -1, 1)$ we tried.

Five shares with equal correlation. Take Lesson 30's five shares, every pair with correlation $\rho = 0.5$. The vector of ones is an eigenvector: each row of the correlation matrix adds up to $1 + 4\rho$. Every vector whose entries add up to zero is an eigenvector with eigenvalue $1 - \rho$ (Exercise 31.4). So the eigenvalues are 3, 0.5, 0.5, 0.5, 0.5:

- the first component, all five shares moving together, the "market", carries 0.60 of the total variance (3 out of 5);
- the four others, ways of moving some shares up and some down, carry 0.5 each.

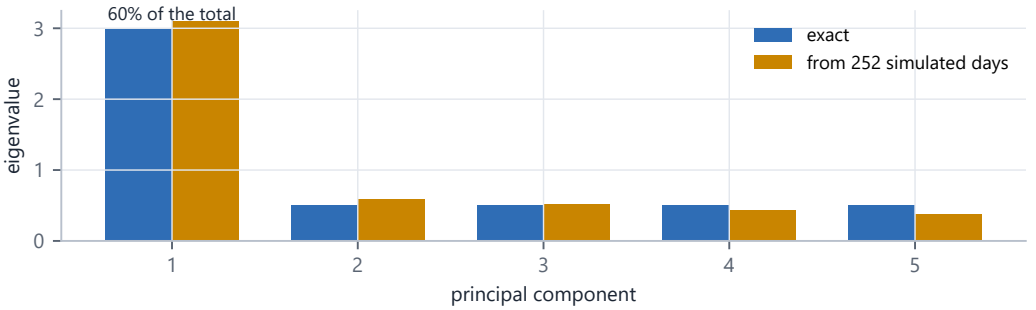


Figure 31.2. Eigenvalues of the five-share correlation matrix with ρ = 0.5 (blue), and of the correlation matrix estimated from 252 simulated daily returns (amber).

Estimated from a year of simulated days, the eigenvalues come out as 3.09, 0.58, 0.52, 0.43 and 0.37. The first component carries 0.62. Sampling noise spreads the small eigenvalues apart: the largest estimates are too big and the smallest too small, a systematic effect worth remembering whenever a correlation matrix is estimated from data.

Repairing an impossible table. Replace the negative eigenvalues by (almost) zero, rebuild $Q\Lambda Q^\top$, and divide rows and columns by the square roots of the diagonal so that it is a correlation matrix again. For our table this gives

$$\begin{pmatrix} 1 & 0.5 & -0.5 \\ 0.5 & 1 & 0.5 \\ -0.5 & 0.5 & 1 \end{pmatrix},$$

whose smallest eigenvalue is 7.9e-07: valid, and only just. The repair halves every correlation, and under the new table the portfolio $(1, -1, 1)$ has zero variance. A repaired table is a different table: price with it only knowing that.

The five-share basket again

The basket's average return is $w^\top R$ with $w = (\frac{1}{5}, \dots, \frac{1}{5})$, so its volatility is $\sqrt{w^\top \Sigma w}$. With equal volatilities and equal correlations this simplifies (Exercise 31.6):

$$\sigma_{\text{basket}} = \sigma\sqrt{\frac{1 + (n-1)\rho}{n}} = 0.2\sqrt{\frac{1 + 4 \times 0.5}{5}} = 0.1549.$$

However many shares you add, it can't fall below $\sigma\sqrt{\rho} = 0.1414$: the part of the risk that every share shares can't be diversified away.

The price. Lesson 30 drew 1,000,000 sets of five correlated normals with numpy's Cholesky. The hand-written one gives the same factor, so with the same random numbers the price is 5.3391, equal to Lesson 30's 5.3391 to every digit. A Black-Scholes call at the basket's volatility gives 5.3301. Lesson 30 used 0.1550 instead of 0.1549, from matching the first two moments of the basket's price rather than of its return; the difference is the lognormal correction of Lesson 5.

What to remember

- **Portfolio variance:** $\text{Var}(w^\top R) = w^\top \Sigma w$, with $\Sigma_{ij} = \text{Cov}(R_i, R_j)$.
- **Valid tables:** every covariance or correlation matrix is positive semi-definite. A table filled in pair by pair may not be.
- **Cholesky:** $\Sigma = LL^\top$ with L lower triangular; $X = LZ$ has covariance Σ. Two shares: $X_2 = \sigma_2(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2)$. It fails exactly when the matrix isn't positive definite.
- **Eigenvalues:** $\Sigma = Q\Lambda Q^\top$; principal components are uncorrelated directions, with the eigenvalues as their variances. Positive semi-definite means no negative eigenvalue.
- **Equal correlation:** eigenvalues $1 + (n-1)\rho$ (the market) and $1 - \rho$; basket volatility $\sigma\sqrt{(1 + (n-1)\rho)/n}$, never below $\sigma\sqrt{\rho}$.

Common mistakes

- **Multiplying the normals by Σ, or by its entries, instead of by L.** ΣZ has covariance Σ^2, not Σ.
- **Using $L^\top Z$ instead of LZ.** Its covariance is $L^\top L$, which is a different matrix.
- **Mixing correlations and covariances.** A Cholesky factor of the correlation matrix gives standardised draws; scale each by its volatility afterwards.

- **Trusting a table because each entry looks reasonable.** Check it: compute its Cholesky factor or its eigenvalues.
- **Reading a small estimated eigenvalue as real.** With limited data the smallest eigenvalues are biased down and the largest up.

Exercises

Solutions are in *Part 5: Solutions*.

31.1 Two shares have volatilities 20% and 30% and correlation 0.5. Find the variance and volatility of a portfolio with half its money in each, term by term. Then find the weights that make the variance as small as possible, and that smallest volatility.

31.2 Show that if Z is a vector of independent standard normals and $X = LZ$, then $\mathrm{Cov}(X) = LL^\top$. Find the Cholesky factor L of the covariance matrix of the two shares in Exercise 31.1 by matching entries, and check $LL^\top$.

31.3 Show that the correlation table with $\rho_{12} = \rho_{23} = 0.9$ and $\rho_{13} = -0.9$ is impossible, using the weights $w = (1, -1, 1)$. Keeping $\rho_{12} = \rho_{23} = 0.9$, for which values of ρ_{13} is the table possible? (Use the determinant.)

31.4 Let C be the $n \times n$ matrix with ones on the diagonal and ρ everywhere else. Show that the vector of ones is an eigenvector with eigenvalue $1 + (n-1)\rho$, and that every vector whose entries add up to zero is an eigenvector with eigenvalue $1 - \rho$. What is the smallest ρ that n shares can all share?

31.5 Find the Cholesky factor of the correlation matrix with $\rho_{12} = 0.5$, $\rho_{13} = 0.3$ and $\rho_{23} = 0.4$ by hand, one column at a time.

31.6 An equal-weighted basket holds n shares, each with volatility σ and every pair with correlation ρ. Show that its volatility is $\sigma\sqrt{(1 + (n-1)\rho)/n}$. Evaluate it for $n = 5$, $\rho = 0.5$, $\sigma = 20\%$, and find its limit as $n \to \infty$. What does the limit say about diversification?

31.7 *(Code)* Write a Cholesky routine that reports which pivot fails when the matrix isn't positive definite. Test it on the matrices of this lesson, and check it against numpy's on 1000 random correlation matrices.

31.8 *(Code)* Price the five-share basket call for $\rho = 0, 0.25, 0.5, 0.75, 1$ by Monte Carlo, and compare each with a Black-Scholes call at the basket's volatility $\sigma\sqrt{(1 + 4\rho)/5}$. Where is the approximation best, and why is it exact at $\rho = 1$?

Self-test

1. Why must every covariance matrix be positive semi-definite?
2. What is the Cholesky factor of the 2 × 2 correlation matrix with correlation ρ?
3. What is the largest eigenvalue of the five-share correlation matrix with $\rho = 0.5$, and what does its eigenvector describe?

Companion notebook: `s5e31_linear_algebra.ipynb`. Every number in this chapter is computed by it.

LESSON 32 · VIDEO 32

Ordinary Differential Equations

First-order linear equations and the integrating factor, applied to the average and the variance of an interest rate; the CIR bond, whose equation has a square in it (a Riccati equation), solved in closed form; Euler's method against Runge-Kutta, and why one is first order and the other fourth; and an equation whose solution runs off to infinity in finite time.

Why this lesson

Lesson 28 priced a bond under the Vasicek model by guessing $P = e^{A(\tau) - B(\tau) r}$ and found that the guess turned a partial differential equation into two **ordinary** differential equations for A and B: equations in one variable, τ. This happens again and again:

- the bond prices of every **affine** interest-rate model (Lessons 39 to 41);
- the **Heston** model's characteristic function (Lesson 48), which comes from the same kind of guess;
- the average and the variance of any linear stochastic process.

Some of these ODEs can be solved by hand with one trick, the **integrating factor**. Others contain a square of the unknown, which makes them **Riccati equations**; those can sometimes be solved by a substitution, and otherwise must be solved numerically. This lesson covers all three, and ends with a warning: a Riccati equation can blow up.

What you need from earlier lessons

- **The growth equation** (Lesson 2). $m' = km$ has solution $m(t) = m(0)e^{kt}$.
- **The Vasicek model and the affine bond** (Lesson 28). $dr = \kappa(\bar{r} - r)\,dt + s\,dW$; the bond's PDE, and $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$.
- **Euler's method** (Lesson 17). Step along the slope at the start of each step.
- **Averages kill dW terms** (Lesson 16). For an Itô process, $\frac{d}{dt}E[X_t]$ is the average of its dt term.

Numbers. Lesson 28's Vasicek model: speed $\kappa = 0.5$, long-run level $\bar{r} = 4\%$, volatility $s = 1\%$, today's rate $r_0 = 3\%$.

The picture

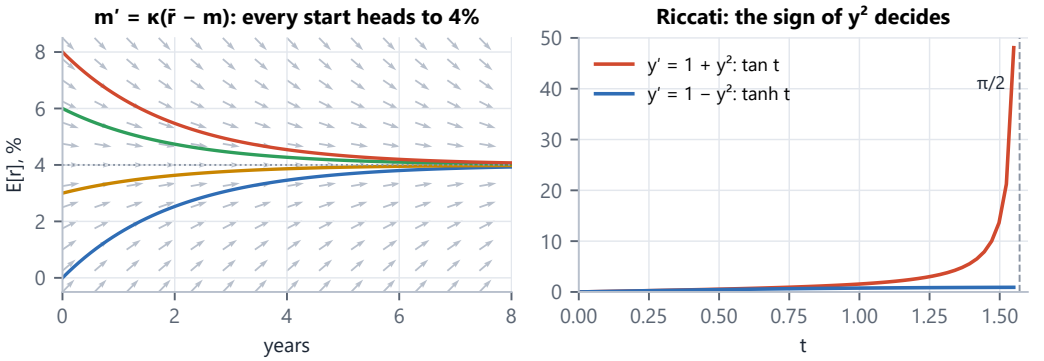


Figure 32.1. Left: the slope field of m′ = κ(r̄ − m), drawn as short arrows, with the solutions from four starting rates; all of them close in on 4%. Right: two Riccati equations that differ only in the sign of the square. With $+y^2$ the solution, tan t, runs off to infinity at t = π/2; with $-y^2$ it, tanh t, settles at 1.

An ODE says, at every point, which way the solution is heading. Solving it means following the arrows. Usually they lead somewhere sensible. With the wrong sign of a square, they lead to infinity in finite time.

The integrating factor

The equation. A **first-order linear ODE** has the form

$$y'(t) + a\,y(t) = b(t),$$

with the unknown y and its derivative appearing only to the first power. For now a is a constant.

The trick. Multiply both sides by e^{at}, the **integrating factor**:

$$e^{at}y' + a\,e^{at}y = e^{at}b(t).$$

Step 1. Recognise the left side. By the product rule (Lesson 1), $\frac{d}{dt}\left(e^{at}y\right) = e^{at}y' + a\,e^{at}y$. So the left side is a single derivative:

$$\frac{d}{dt}\left(e^{at}y\right) = e^{at}b(t).$$

Step 2. Integrate from 0 to t.

$$e^{at}y(t) - y(0) = \int_0^t e^{au} b(u)\,du.$$

Step 3. Divide by e^{at}.

$$\boxed{y(t) = y(0)e^{-at} + \int_0^t e^{-a(t-u)} b(u)\,du}$$

When b is a constant, the integral is $\frac{b}{a}(1 - e^{-at})$, and

$$y(t) = \frac{b}{a} + \Big(y(0) - \frac{b}{a}\Big)e^{-at}.$$

The solution forgets its starting value at the rate a and settles at b/a, the level where $y' = 0$.

The average of the Vasicek rate. Take expectations of $dr = \kappa(\bar r - r)\,dt + s\,dW$. The dW term has mean zero, so $m(t) = E[r_t]$ solves $m' = \kappa(\bar r - m)$, that is, $m' + \kappa m = \kappa \bar r$. Here $a = \kappa$ and $b = \kappa\bar r$, so $b/a = \bar r$:

$$E[r_t] = \bar r + (r_0 - \bar r)e^{-\kappa t}.$$

Half the gap to $\bar r$ closes in $\ln 2/\kappa = 1.386$ years. These are the curves in the left of the picture.

Its variance. Write $D_t = r_t - m(t)$, the rate's distance from its average. Then $dD = -\kappa D\,dt + s\,dW$. By Itô's lemma (Lesson 16),

$$d(D^2) = 2D\,dD + (dD)^2 = (-2\kappa D^2 + s^2)\,dt + 2sD\,dW.$$

Take expectations: $v(t) = E[D_t^2]$, the variance, solves $v' + 2\kappa v = s^2$, with $v(0) = 0$. The same formula, with $a = 2\kappa$ and $b = s^2$:

$$\mathrm{Var}(r_t) = \frac{s^2}{2\kappa}\big(1 - e^{-2\kappa t}\big).$$

In the long run the rate's standard deviation is $s/\sqrt{2\kappa} = 0.0100$: the pull back to $\bar r$ stops it wandering off, as it would without the pull. Checked against 20,000 simulated paths:

The Vasicek rate's average and standard deviation, from the integrating factor and from simulation (daily steps).

Time (years)	Mean of r (formula)	Mean of r (simulated)	sd of r (formula)	sd of r (simulated)
0.5	0.03221	0.03221	0.00627	0.00630
1.0	0.03393	0.03398	0.00795	0.00789
2.0	0.03632	0.03629	0.00930	0.00931
5.0	0.03918	0.03919	0.00997	0.00992

The bond again. In Lesson 28, matching the terms in r gave $B' = 1 - \kappa B$ with $B(0) = 0$. That is $B' + \kappa B = 1$, so

$$B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}.$$

$B(1) = 0.7869$, $B(5) = 1.8358$, $B(10) = 1.9865$, and B never exceeds $1/\kappa = 2$. B measures how much the bond's log price moves when today's rate moves, so a 30-year bond is barely more sensitive to today's rate than a 10-year one: any shock to the rate fades before the bond matures.

Riccati equations and the CIR bond

The CIR model. In Vasicek, the rate is normal and can go negative. The **CIR model** (Cox, Ingersoll and Ross) makes the volatility shrink as the rate falls:

$$dr = \kappa(\bar{r} - r)\,dt + s_c\sqrt{r}\,dW.$$

To compare like with like, choose s_c so that today's volatility is Vasicek's: $s_c\sqrt{r_0} = 1\%$, so $s_c = 0.0577$.

The bond's equation. Feynman-Kac with a random discount rate (Lesson 28) gives, for $P(t, r)$,

$$P_t + \kappa(\bar{r} - r)P_r + \tfrac{1}{2}s_c^2 r\,P_{rr} = rP.$$

Step 1. Try the affine guess $P = e^{A(\tau) - B(\tau)r}$ with $\tau = T - t$. Then $P_t = -(A' - B'r)P$, $P_r = -BP$ and $P_{rr} = B^2P$.

Step 2. Substitute and divide by P.

$$-A' + B'r - \kappa(\bar{r} - r)B + \tfrac{1}{2}s_c^2 B^2\,r = r.$$

Step 3. Collect. This must hold for every r, so the terms with r and the terms without must each balance:

$$B' = 1 - \kappa B - \tfrac{1}{2} s_c^2 B^2, \qquad A' = -\kappa \bar{r} B, \qquad A(0) = B(0) = 0.$$

The square comes from the volatility $s_c\sqrt{r}$: its square, $s_c^2 r$, multiplies $P_{rr} = B^2 P$ and so lands in the equation for the coefficient of r. An equation $y' = a + by + cy^2$ is a **Riccati equation**. It isn't linear, and the integrating factor doesn't apply.

Step 4. The substitution. Write $B = 2u'/(s_c^2 u)$ for a new unknown u. The square cancels, and u solves the *linear* second-order equation (Exercise 32.4)

$$u'' + \kappa u' - \tfrac{1}{2} s_c^2 u = 0.$$

Try $u = e^{\lambda \tau}$: $\lambda^2 + \kappa\lambda - \tfrac{1}{2}s_c^2 = 0$, so $\lambda = \tfrac{1}{2}(-\kappa \pm \gamma)$ with $\gamma = \sqrt{\kappa^2 + 2s_c^2} = 0.5066$. Combining the two exponentials so that $B(0) = 0$ gives

$$B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}, \qquad A(\tau) = \frac{2\kappa\bar{r}}{s_c^2} \ln \frac{2\gamma\, e^{(\gamma+\kappa)\tau/2}}{(\gamma + \kappa)(e^{\gamma\tau} - 1) + 2\gamma}.$$

As $\tau \to \infty$, B settles where its right-hand side is zero, at $(\gamma - \kappa)/s_c^2 = 1.9868$, a little below Vasicek's 2.

The two models with the same volatility today. Their yields differ by less than a basis point out to 10 years.

Maturity (years)	B (Vasicek)	B (CIR)	Yield (Vasicek, %)	Yield (CIR, %)
1	0.7869	0.7866	3.2119	3.2118
2	1.2642	1.2625	3.3645	3.3643
5	1.8358	1.8281	3.6235	3.6221
10	1.9865	1.9742	3.7873	3.7840
30	2.0000	1.9868	3.9153	3.9101

Why the long yields differ. As the rate rises towards 4%, CIR's volatility $s_c\sqrt{r}$ rises with it, to about 1.15%. More volatility means more convexity in the bond, which lowers its yield. At 30 years, CIR's yield is 3.910% against Vasicek's 3.915%.

Checked by simulation. 50,000 paths of the CIR rate (Euler steps of 1/100 year, with the rate floored at zero inside the square root) price the 5-year bond at 0.8345, with a standard error of 0.0001. The formula gives 0.8343.

Solving numerically: Euler and Runge-Kutta

Most ODEs have no closed form. Then we step.

Euler's method. For $y' = f(y)$, take $y_{n+1} = y_n + h\,f(y_n)$: follow the slope at the start of the step for a time h. By Taylor's theorem, $y(t+h) = y(t) + hy'(t) + \frac{1}{2}h^2y''(t) + \ldots$, so each step makes an error of about $\frac{1}{2}h^2y''$. There are T/h steps, so the errors add up to something proportional to h: Euler is **first order**. Halve the step, halve the error.

Runge-Kutta 4. Take four slopes inside each step:

$$k_1 = f(y_n), \quad k_2 = f\big(y_n + \tfrac{h}{2}k_1\big), \quad k_3 = f\big(y_n + \tfrac{h}{2}k_2\big), \quad k_4 = f(y_n + hk_3),$$

$$y_{n+1} = y_n + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

k_1 is the slope at the start, k_2 and k_3 two estimates of the slope in the middle, and k_4 the slope at the end. The weights 1, 2, 2, 1 are Simpson's rule for averaging a slope across the step. Matching the Taylor series term by term (a long but routine calculation) shows that each step's error is of order h^5, so the total is of order h^4: **fourth order**. Halve the step and the error falls by 16.

Runge-Kutta 4 for a system of equations: y holds A and B together.

```
def rk4(f, y0, t_end, h):
    y = np.array(y0, float)
    for _ in range(round(t_end / h)):
        k1 = f(y); k2 = f(y + h / 2 * k1); k3 = f(y + h / 2 * k2); k4 = f(y + h * k3)
        y = y + h / 6 * (k1 + 2 * k2 + 2 * k3 + k4)
    return y
```

The test. Solve the CIR pair (A, B) from $\tau = 0$ to 10 with each method, and compare $B(10)$ with the closed form, 1.974230:

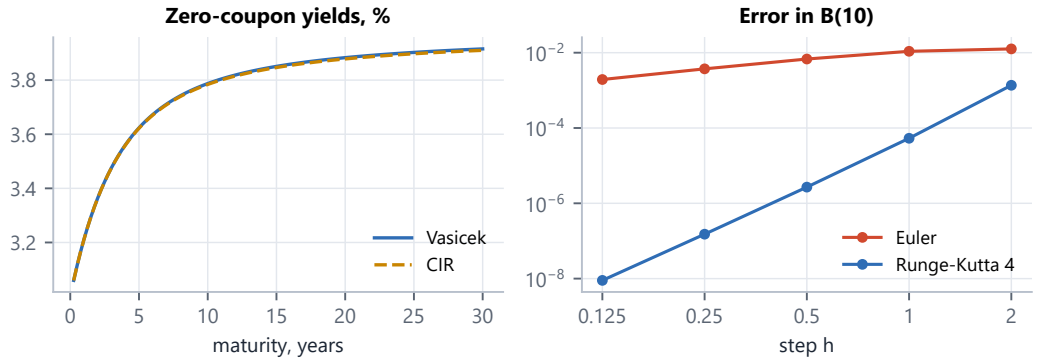


Figure 32.2. Left: CIR and Vasicek zero-coupon yields with the same volatility today. Right: the error in B(10) against the step, on log scales. Euler's line has slope 1; Runge-Kutta's has slope 4.

Errors in B(10) for the CIR Riccati equation.

Step (h)	STEPS	EULER ERROR	RUNGE-KUTTA 4 ERROR
2.000	5	1.3e-02	-1.4e-03
1.000	10	1.1e-02	-5.3e-05
0.500	20	6.8e-03	-2.7e-06
0.250	40	3.7e-03	-1.5e-07
0.125	80	1.9e-03	-9.0e-09

From the last two rows, halving h divides Euler's error by 1.92 and Runge-Kutta's by 16.9: first order and fourth order. Runge-Kutta costs four slope evaluations per step instead of one, but with 20 steps it is already more accurate than Euler with 80, by a factor of thousands. With $h = 0.5$, its 10-year bond price is 0.6849558605, against 0.6849558604 from the formula.

When the solution blows up

Two equations. Solve $y' = 1 + y^2$ with $y(0) = 0$. Separate the variables: $dy/(1 + y^2) = dt$, so $\arctan y = t$ and $y = \tan t$. As t approaches $\pi/2 \approx 1.5708$, $\tan t$ goes to infinity: **the solution blows up in finite time**. Change the sign of the square, $y' = 1 - y^2$, and the solution is $\tanh t$, which rises smoothly to 1 and stays there.

Why it matters. The CIR equation for B has $-\frac{1}{2}s_c^2 B^2$: the safe sign. But the same machinery, applied to other quantities (moments of the integrated rate, or the moment generating function of the log price in the Heston model, Lesson 48), produces Riccati equations with the dangerous sign. Then a moment or a price can become infinite at a finite maturity, and a fixed-step solver won't tell you.

A fixed-step solver doesn't notice. Runge-Kutta with $h = 0.01$ on $y' = 1 + y^2$ just returns larger and larger numbers. A simple check is **step doubling**: take each step once with h, and again as two steps of $h/2$. The difference estimates the error made in that step.

Runge-Kutta with h = 0.01 on y′ = 1 + y². The error estimate climbs by orders of magnitude before the numbers themselves look wrong.

Time (t)	RUNGE-KUTTA 4	tan(t)	STEP-DOUBLING ESTIMATE
0.50	0.546	0.546	1.4e-12
1.00	1.557	1.557	1.8e-11
1.50	14.101	14.101	2.3e-05
1.55	48.076	48.078	1.7e-02
1.56	92.573	92.620	3.7e-01
1.57	868.654	1255.766	2.8e+02

At $t = 1.5$ the solver is still right to six digits, but its estimate has jumped from 10^{-11} to 10^{-5}. By 1.57, just short of $\pi/2 = 1.5708$, the answer is off by a third. A solver that shrinks its step when the estimate grows would stall at the blow-up and report it; one with a fixed step marches straight through.

What to remember

- **Integrating factor:** for $y' + ay = b(t)$, multiply by e^{at}; then $y(t) = y(0)e^{-at} + \int_0^t e^{-a(t-u)}b(u)\,du$. With constant b, $y \to b/a$ at the rate a.
- **Vasicek:** $E[r_t] = \bar{r} + (r_0 - \bar{r})e^{-\kappa t}$; $\mathrm{Var}(r_t) = \frac{s^2}{2\kappa}(1 - e^{-2\kappa t})$; $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$.
- **Riccati:** $y' = a + by + cy^2$. A volatility proportional to $\sqrt{r}$ puts a B^2 in the bond's equation. The substitution $y = -u'/(cu)$ makes it linear.
- **CIR bond:** $B' = 1 - \kappa B - \frac{1}{2}s_c^2 B^2$, $A' = -\kappa\bar{r}B$, closed form with $\gamma = \sqrt{\kappa^2 + 2s_c^2}$.
- **Euler:** error $\propto h$. **Runge-Kutta 4:** error $\propto h^4$.
- **Blow-up:** with the wrong sign of the square, a Riccati solution can reach infinity in finite time. Step doubling detects it.

Common mistakes

- **Putting the bond's starting condition at the wrong end.** A and B are zero at $\tau = 0$, which is maturity, not today.
- **Getting the sign of a wrong in the integrating factor.** For $y' + ay = b$ it is e^{+at}.
- **Forgetting the s^2 in the variance equation.** It comes from $(dD)^2$, Itô's extra term.
- **Treating a Riccati equation as linear.** The B^2 term matters, and changes the long-run value of B.
- **Trusting a fixed-step solver near a blow-up.** It gives large, wrong, finite numbers without complaint.

Exercises

Solutions are in *Part 5: Solutions*.

32.1 Solve $m' = \kappa(\bar{r} - m)$ with $m(0) = r_0$ by the integrating factor $e^{\kappa t}$. With Lesson 28's numbers ($\kappa = 0.5$, $\bar{r} = 4\%$, $r_0 = 3\%$), find $E[r_1]$ and $E[r_5]$, and the time it takes to close half the gap to $\bar{r}$.

32.2 The variance $v(t)$ of the Vasicek rate solves $v' = -2\kappa v + s^2$ with $v(0) = 0$. Solve it, find the long-run standard deviation, and evaluate the standard deviation at $t = 1$ with $s = 1\%$.

32.3 For the Vasicek bond $P = e^{A(\tau) - B(\tau) r}$, show that $B' = 1 - \kappa B$ and $A' = -\kappa\bar{r}B + \frac{1}{2}s^2 B^2$, with $A(0) = B(0) = 0$. Solve for B with an integrating factor, and find its limit as $\tau \to \infty$.

32.4 Show that the substitution $y = -u'/(cu)$ turns the Riccati equation $y' = a + by + cy^2$ into the linear equation $u'' - bu' + acu = 0$. For the CIR equation $B' = 1 - \kappa B - \frac{1}{2}s_c^2 B^2$, find the two exponents of u, and the limit of B as $\tau \to \infty$.

32.5 Solve $y' = y^2$ with $y(0) = 1$, and $y' = 1 + y^2$ with $y(0) = 0$. When does each blow up? Then solve $y' = 1 - y^2$ with $y(0) = 0$, and explain why the sign of the square decides whether the solution stays finite.

32.6 Apply Euler's method with step h to $y' = -\kappa y$, $y(0) = 1$. Show that $y_n = (1 - \kappa h)^n$, and compare y_{10} with $e^{-0.5}$ for $\kappa = 0.5$, $h = 0.1$. Show that the error at a fixed time is roughly proportional to h. What happens when $h > 2/\kappa$?

32.7 *(Code)* Use Runge-Kutta 4 with $h = 0.1$ to compute the CIR yield curve from 1 to 30 years, and compare it with the closed form. What is the largest error, in basis points?

32.8 *(Code)* Write a step-doubling error estimate for Runge-Kutta 4, apply it to $y' = 1 + y^2$ from $t = 0$ to $t = 1.57$ with $h = 0.01$, and find where the estimate first exceeds 10^{-6}. How close is that to the blow-up time?

Self-test

1. What is the integrating factor for $y' + 3y = 1$, and what is the solution with $y(0) = 0$?
2. Why does the CIR bond's equation for B contain B^2, while Vasicek's doesn't?
3. You halve the step of Runge-Kutta 4. Roughly how much smaller does the error get?

Companion notebook: `s5e32_odes.ipynb`. Every number in this chapter is computed by it.

LESSON 33 · VIDEO 33

Complex Numbers and Fourier Transforms

Complex numbers from scratch, and Euler's formula from the Taylor series; the characteristic function, which every random variable has; turning it back into a density, and into the probabilities in the Black-Scholes formula, by Fourier inversion; the course's call priced without a density; and the complex-logarithm trap that waits in the Heston model.

Why this lesson

For the Black-Scholes model we know the density of the share price at expiry, and pricing is an integral against it. For most models beyond Black-Scholes we don't. The **Heston** model (Lesson 47), where the volatility itself moves, has no formula for its density. But it does have a formula for something closely related: its **characteristic function**, $\varphi(u) = E[e^{iuX}]$. And a characteristic function determines the distribution: from it, by **Fourier inversion**, we can get back the density, the probability of finishing in the money, and the price.

That needs complex numbers, which the course hasn't used yet. This lesson builds them from the start, then the characteristic function, then the inversion. To make sure the machinery works, we use it where we already know the answer: the Black-Scholes call, 7.1281. Lesson 48 will point the same code at Heston.

What you need from earlier lessons

- **Taylor series** (Lesson 1). $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$, and the series for $\cos x$ and $\sin x$.
- **The moment generating function and completing the square** (Lesson 4). $E[e^{tX}] = e^{tm + \frac{1}{2}s^2t^2}$ for $X \sim N(m, s^2)$.
- **The call as two probabilities** (Lessons 20 and 27). $C = S_0 N(d_+) - Ke^{-rT}N(d_-)$, where $N(d_-)$ is the risk-neutral chance of finishing in the money and $N(d_+)$ the same chance in share units.

The picture

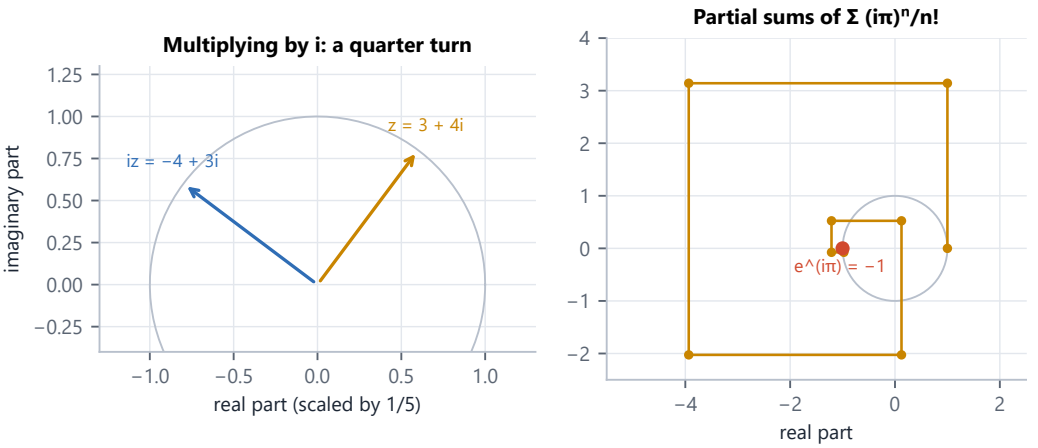


Figure 33.1. Left: a complex number is a point in the plane; multiplying by i turns it a quarter turn about 0 without changing its length. Right: the partial sums of 1 + iπ + (iπ)²/2! + ⋯ spiral in to e^(iπ) = −1.

Complex numbers

Definition. The number i is defined by $i^2 = -1$. A **complex number** is $z = a + bi$, with a (the **real part**) and b (the **imaginary part**) ordinary numbers. Complex numbers are added and multiplied like ordinary algebra, replacing i^2 by -1 wherever it appears.

Example. $(3 + 4i)(1 - 2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$.

The plane. Draw $a + bi$ as the point (a, b). Its distance from 0 is the **modulus** $|z| = \sqrt{a^2 + b^2}$, and the angle from the positive real axis is the **argument**. For $z = 3 + 4i$, $|z| = 5$ and the argument is 53.13°.

Division. The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$, and $z\bar{z} = a^2 + b^2 = |z|^2$, an ordinary number. So $1/z = \bar{z}/|z|^2$. For $z = 3 + 4i$: $1/z = (3 - 4i)/25 = 0.12 - 0.16i$.

Multiplying by i turns a quarter turn. $i(a + bi) = -b + ai$: the point (a, b) goes to $(-b, a)$, the same distance from 0, a quarter turn anticlockwise. For $z = 3 + 4i$, $iz = -4 + 3i$ (the left of the picture).

Euler's formula

What could $e^{i\theta}$ mean? Use the Taylor series of e^x (Lesson 1) with $x = i\theta$, and the powers of i: $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and then round again.

Step 1. Expand.

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \cdots$$

Step 2. Separate the real and imaginary terms.

$$e^{i\theta} = \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\Big).$$

Step 3. Recognise the two series as those of $\cos\theta$ and $\sin\theta$:

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

What follows. $|e^{i\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$: $e^{i\theta}$ is the point on the unit circle at angle θ. Every complex number can be written $z = |z|e^{i\theta}$, and then multiplication is simple: **multiply the moduli and add the angles**, since $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$. With $\theta = \pi$, $e^{i\pi} = -1$. The right of the picture follows the series: after 8 terms it is at the point (-0.9760, -0.0752), and after 20 within 5.3e-10 of -1.

The characteristic function

Definition. The **characteristic function** of a random variable X is

$$\varphi(u) = E[e^{iuX}] = E[\cos uX] + i\,E[\sin uX], \qquad u \text{ a real number.}$$

It always exists. $|e^{iuX}| = 1$, so the average of something no bigger than 1 is finite. The moment generating function $E[e^{tX}]$ needn't be: for the share price itself, $E[e^{tS_T}]$ is infinite for every $t > 0$ (Exercise 33.4). Three properties follow straight from the definition: $\varphi(0) = 1$, $|\varphi(u)| \le 1$, and $\varphi(-u) = \overline{\varphi(u)}$.

For a normal. If $X \sim N(m, s^2)$, the completing-the-square calculation of Lesson 4 goes through with iu in place of a real t (Exercise 33.3):

$$\varphi(u) = e^{ium - \frac{1}{2}s^2u^2}.$$

For a standard normal, $\varphi(u) = e^{-u^2/2}$, a real number, because the density is symmetric and the sine parts cancel. A million simulated draws agree:

The characteristic function of a standard normal, by simulation.

Frequency (u)	E[cos(uZ)], simulated	E[sin(uZ)], simulated	e^(−u²/2)
0.5	0.8825	-0.0001	0.8825
1.0	0.6066	-0.0005	0.6065
2.0	0.1362	-0.0008	0.1353

For the log price. Under the risk-neutral probability, $X = \ln S_T$ is normal with mean $m = \ln S_0 + (r - \frac{1}{2}\sigma^2)T = 4.61517$ and variance $v = \sigma^2 T = 0.04$. So

$$\varphi(u) = \exp\left(iu\,m - \tfrac{1}{2}v\,u^2\right).$$

Why bother. Two reasons. First, for independent X and Y, $E[e^{iu(X+Y)}] = E[e^{iuX}]E[e^{iuY}]$: characteristic functions of sums multiply, which is how models built from independent pieces get their formulas. Second, for many models (Heston, jumps, affine rates) the characteristic function has a formula even though the density hasn't.

From characteristic function to density

Fourier inversion (stated here, and checked numerically). A characteristic function determines its density:

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-iux}\varphi(u)\,du = \frac{1}{\pi}\int_{0}^{\infty} \mathrm{Re}\left[e^{-iux}\varphi(u)\right]du.$$

The second form uses $\varphi(-u) = \overline{\varphi(u)}$: the parts for $-u$ and u are conjugates, and their imaginary parts cancel. The idea behind the formula: $e^{-iux}\varphi(u) = E[e^{iu(X-x)}]$, and averaging $e^{iu(X-x)}$ over all frequencies u cancels out everywhere except where X is near x.

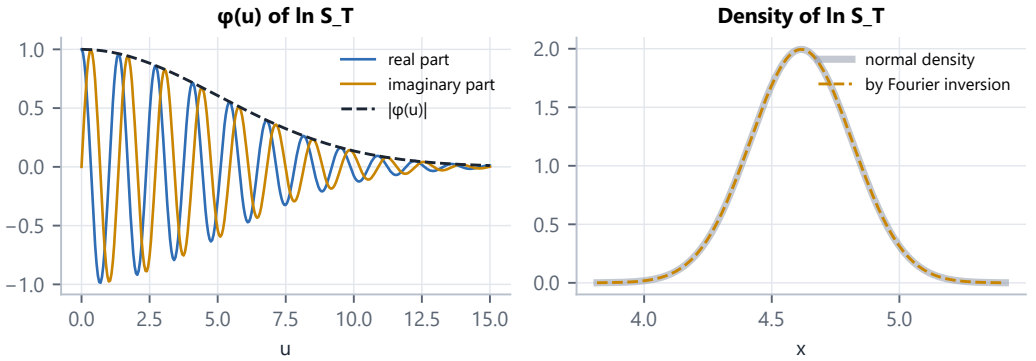


Figure 33.2. Left: the characteristic function of ln S_T. It spins round at a rate set by the mean (about 4.6) while its size, e^(−0.02u²), shrinks. Right: the density of ln S_T, and the same density rebuilt from φ by Fourier inversion.

In numbers. Approximate the integral by a midpoint sum with step $du = 0.01$ up to $u = 60$. At 161 points across the density, the largest difference from the normal density is 4.4e-16, against a peak of 1.995.

Pricing the call by Fourier inversion

The call needs two probabilities of the same event, $S_T > K$, one in each unit (Lesson 27). The **Gil-Pelaez formula** (stated) turns a characteristic function into such a probability, for $X = \ln S_T$ and $k = \ln K$:

$$P(X > k) = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\Big[\frac{e^{-iuk}\varphi(u)}{iu}\Big]du$$

The $1/(iu)$ looks dangerous at $u = 0$, but the bracket tends to the finite value $E[X] - k$ there (Exercise 33.6), here -0.03879. A midpoint sum never evaluates $u = 0$ anyway.

The strike term. With the risk-neutral φ: $\tilde{P}(S_T > K) = 0.42311$. The formula's $N(d_-)$ is 0.42311.

The share term. Lesson 27's share measure weights each outcome by $e^X/E[e^X]$. Its characteristic function is therefore

$$\varphi_S(u) = \frac{E[e^{iuX}e^X]}{E[e^X]} = \frac{\varphi(u-i)}{\varphi(-i)}$$

(since $e^{i(u-i)X} = e^{iuX}e^X$). The denominator is $E[S_T] = 103.0455$, the forward price. Gil-Pelaez with φ_S gives $P^S(S_T > K) = 0.50241$, against $N(d_+) = 0.50241$.

The call. $C = S_0P^S(S_T > K) - Ke^{-rT}\tilde{P}(S_T > K) =$ 7.1281. The formula gives 7.1281: they differ by 7.1e-15. No density was used.

Gil-Pelaez by a midpoint sum.

```
def gil_pelaez(phi_, k, U=60.0, du=0.01):
    u = np.arange(du / 2, U, du)
    return 0.5 + (np.exp(-1j * u * k) * phi_(u) / (1j * u)).real.sum() * du / math.pi
```

Two numerical choices. Where to stop the integral (U), and how big a step to take (du).

Stopping too early (du = 0.01). The integrand shrinks like e^(−0.02u²), and the price is right to 4 decimals only from about U = 20.

Upper limit (U)	CALL	ERROR
2	1.576601	-5.6e+00
5	4.589064	-2.5e+00
10	6.771120	-3.6e-01
20	7.127604	-4.6e-04

Large steps (U = 60). With fifteen points the price is still exact to 12 decimals.

Step (du)	POINTS UP TO 60	2π/du	CALL	ERROR
10	6	0.628	7.004826	-1.2e-01
8	7	0.785	7.120132	-7.9e-03
6	10	1.047	7.128044	-2.1e-05
4	15	1.571	7.128065	-9.4e-13
2	30	3.142	7.128065	-7.1e-15

Why such big steps work. A sum over $u = du/2, 3du/2, \ldots$ can't tell x from $x + 2\pi/du$: the terms e^{-iux} are the same for both (this is called **aliasing**). So a step du effectively folds the distribution of $\ln S_T$ onto a range of width $2\pi/du$. With $du = 4$ that range is 1.57 in log price, about 8 standard deviations, where the density is negligible: no harm done. With $du = 10$ it is 0.63, about 3 standard deviations, and probability that has been folded over spoils the answer. The step must be small enough for $2\pi/du$ to cover the distribution; beyond that, extra points buy nothing.

The complex-log trap

Logarithms of complex numbers. Write $z = |z|e^{i\theta}$. Then $\ln z = \ln|z| + i\theta$, but θ is only defined up to a whole turn: $\theta + 2\pi$ is the same direction. So $\ln z$ has many values, $\ln|z| + i(\theta + 2\pi n)$. Computers return the **principal** one, with θ in $(-\pi, \pi]$.

Powers. A power is computed as $z^c = e^{c \ln z}$. When c is not a whole number, the different logarithms give different answers, and the principal one **jumps** whenever z crosses the negative real axis, where its angle flips from π to $-\pi$.

A toy. Let $z(u) = e^{iu} + 0.3$, which circles the origin as u runs on, and compute $\sqrt{z(u)}$. The true function changes smoothly with u. Computed with the principal logarithm, it jumps 3 times on $0 \le u \le 6\pi$, flipping sign each time z crosses the negative axis. Keeping track of the angle as it winds (**unwrapping** it, adding 2π whenever it would jump) removes every jump: the largest change between neighbouring points is then 0.0094.

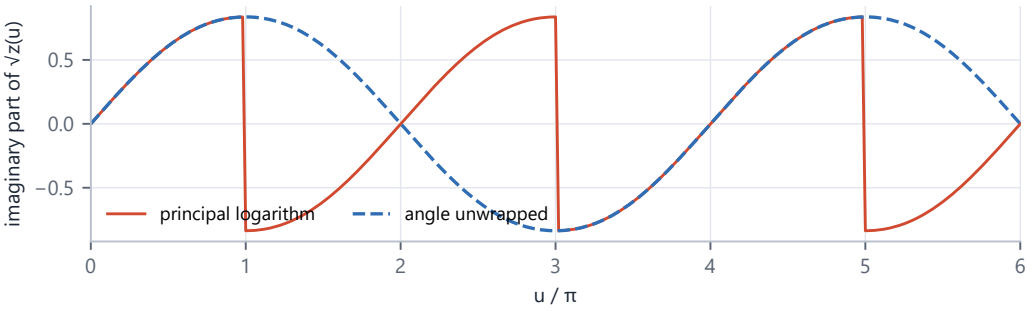


Figure 33.3. The imaginary part of √z(u) for z(u) = e^(iu) + 0.3. Red: the principal logarithm, which jumps at u = π, 3π and 5π. Blue: the angle unwrapped.

Why it matters. The Heston characteristic function contains exactly such a power of a complex number that circles the origin for long maturities. Integrate a function with jumps like these, and the price is simply wrong, with nothing to warn you. Lesson 48 shows the standard rewriting that avoids it.

What to remember

- **Complex numbers:** $i^2 = -1$; $z = a + bi = |z|e^{i\theta}$; multiplying multiplies moduli and adds angles; $1/z = \bar{z}/|z|^2$.
- **Euler:** $e^{i\theta} = \cos\theta + i\sin\theta$, from the Taylor series; $|e^{i\theta}| = 1$.
- **Characteristic function:** $\varphi(u) = E[e^{iuX}]$; always exists; $\varphi(0) = 1$, $|\varphi| \le 1$, $\varphi(-u) = \overline{\varphi(u)}$. Normal: $e^{ium - \frac{1}{2}s^2u^2}$.
- **Inversion:** $f(x) = \frac{1}{\pi}\int_0^\infty \mathrm{Re}[e^{-iux}\varphi(u)]\,du$; Gil-Pelaez for $P(X > k)$.
- **Pricing:** $C = S_0 P^S(S_T > K) - Ke^{-rT}\tilde{P}(S_T > K)$, with $\varphi_S(u) = \varphi(u - i)/\varphi(-i)$.
- **Numerics:** stop where $|\varphi|$ is negligible; keep $2\pi/du$ wider than the distribution.
- **Branch trap:** complex powers via the principal log jump; unwrap the angle.

Common mistakes

- **Using the moment generating function where it doesn't exist.** The characteristic function always exists.
- **Forgetting the real part.** The probability formulas integrate $\mathrm{Re}[\cdot]$; the imaginary parts cancel between u and $-u$.
- **Stopping the integral where φ is still large.** With a small volatility or a short maturity, φ decays slowly and U must be larger.

- **Computing the share-measure probability with φ instead of $\varphi(u-i)/\varphi(-i)$.** That gives $N(d_-)$ twice.
- **Trusting `log` or `**` on complex numbers along a path.** Check for jumps.

Exercises

Solutions are in *Part 5: Solutions*.

33.1 Let $z = 3 + 4i$. Compute $z(1-2i)$, $|z|$ and $1/z$. Show that multiplying any z by i turns it a quarter turn anticlockwise, without changing its length.

33.2 Derive Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ from the Taylor series of e^x, $\cos x$ and $\sin x$. Deduce that $|e^{i\theta}| = 1$, and use $e^{i(a+b)} = e^{ia}e^{ib}$ to get the addition formulas for $\cos(a+b)$ and $\sin(a+b)$.

33.3 Show that the characteristic function of $N(m, s^2)$ is $e^{ium - \frac{1}{2}s^2u^2}$, by the completing-the-square calculation of Lesson 4 with iu in place of a real number. Write down the characteristic function of $\ln S_T$ under the risk-neutral probability, with the course's numbers.

33.4 Show that every characteristic function has $\varphi(0) = 1$, $|\varphi(u)| \le 1$ and $\varphi(-u) = \overline{\varphi(u)}$. Explain why S_T has no moment generating function ($E[e^{tS_T}]$ is infinite for every $t > 0$), although $\ln S_T$ has a characteristic function.

33.5 Lesson 27's share measure weights each outcome by $e^X / E[e^X]$, with $X = \ln S_T$. Show that the characteristic function of X under it is $\varphi(u-i)/\varphi(-i)$. For the Black-Scholes φ, show that this is a normal characteristic function, and find its mean.

33.6 Show that the Gil-Pelaez integrand $\operatorname{Re}\big[e^{-iuk}\varphi(u)/(iu)\big]$ tends to $E[X] - k$ as $u \to 0$, so the integral has no problem at 0. Evaluate this limit for $X = \ln S_T$ and $k = \ln K$ with the course's numbers.

33.7 *(Code)* Price the call for strikes 80 to 130 in steps of 5 by Gil-Pelaez, and compare with the Black-Scholes formula. Find an upper limit U and a step du that give four correct decimals at every strike.

33.8 *(Code)* For $z(u) = e^{iu} + 0.3$ on $0 \le u \le 6\pi$, compute $z^{1/2}$ with the principal logarithm and with an unwrapped angle. Count the jumps in each, and check that the unwrapped version changes smoothly. What would go wrong in a price that integrates such a function?

Self-test

1. What is $|e^{i\theta}|$, and why?
2. What is the characteristic function of a standard normal?
3. Why do models such as Heston get priced through their characteristic function rather than their density?

Companion notebook: `s5e33_fourier.ipynb`. Every number in this chapter is computed by it.

LESSON 34 · VIDEO 34

Numerical Methods: Roots, Interpolation, Integration, Fitting

Implied volatility by bisection, Newton's method and Brent's method, and the deep out-of-the-money case where Newton fails; forward rates from a few zero rates, by straight lines, a cubic spline and a monotone cubic, and the marks each leaves; Gaussian quadrature, and why a kink spoils it; and least squares, linear and nonlinear, with Levenberg-Marquardt rescuing a fit that Gauss-Newton loses.

Why this lesson

Most numerical work in finance comes down to four questions:

- **Solve for:** which volatility gives this price? (implied volatility, Lesson 50; calibration, Lesson 63)
- **Interpolate between:** what is the zero rate at 7 years, given quotes at 5 and 10? (curve building, Lesson 57)
- **Integrate:** what is the average payoff against this density? (Lesson 20, and every semi-analytic formula)
- **Fit to:** which few parameters best describe a whole smile or a whole curve? (Lessons 48 and 50)

Each has a simple method that works most of the time, and a better one for when it doesn't. This lesson builds both for each question, on the course's own numbers, and shows a case where the simple method fails.

What you need from earlier lessons

- **The call and its vega** (Lessons 20 and 21). $C(\sigma)$ rises with σ, at the rate $\mathcal{V} = S\varphi(d_+)\sqrt{T}$.
- **Taylor's theorem** (Lesson 1). $g(x+\delta) \approx g(x) + g'(x)\delta$.
- **The call as an integral** (Lesson 20, Exercise 20.7). $C = e^{-rT}\int \max(S_T(z) - K, 0)\,\varphi(z)\,dz$.

- **The tridiagonal solver** (Lesson 29). The Thomas algorithm solves a tridiagonal system in about $3n$ operations.

The picture

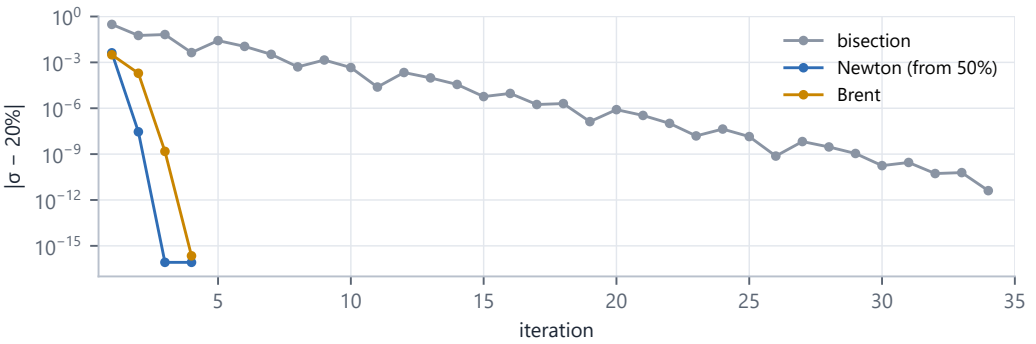


Figure 34.1. How far each method is from σ = 20% after each iteration, on a log scale, when solving C(σ) = 7.1281. Bisection gains a little under a digit every three steps; Newton doubles its correct digits each step; Brent gets close to Newton's speed while keeping bisection's safety.

Root finding: implied volatility

The problem. Given a call price $C^* = 7.1281$, find σ with $g(\sigma) = C(\sigma) - C^* = 0$. Since C rises with σ (vega is positive), there is exactly one answer, and we know it is 20%.

Bisection. Start with a **bracket** $[a, b]$ where g changes sign: $g(0.01) < 0 < g(1)$. Look at the midpoint. Keep the half where the sign still changes. After n halvings the bracket has width $(b - a)/2^n$. It can't fail, but it gains only one binary digit per step: from width 0.99 to 10^{-10} takes 34 steps.

Newton's method. Replace g by its tangent line and solve that instead. From Taylor's theorem, $g(\sigma + \delta) \approx g(\sigma) + g'(\sigma)\delta$, which is zero when $\delta = -g(\sigma)/g'(\sigma)$. Here $g' = \mathcal{V}$, so

$$\sigma_{n+1} = \sigma_n - \frac{C(\sigma_n) - C^*}{\mathcal{V}(\sigma_n)}.$$

From $\sigma_0 = 50\%$ the iterates are 0.19578151, 0.1999999712, 0.200000000000.

Why so fast. Write $e_n = \sigma_n - 0.2$ for the error. Taylor's theorem to second order shows that $e_{n+1} \approx \frac{g''}{2g'} e_n^2$: each error is about the *square* of the one before, so the number of correct digits roughly doubles each step. This is **quadratic convergence**.

When Newton fails. Take a strike of 200, with the true volatility still 20%. The call is worth only 0.003325. Start Newton at 5%: at that volatility the option is so far out of the money that its vega is 3.5e-37, essentially zero. The tangent is almost flat, and the step $-g/g'$ is enormous: the next iterate is 9.4e+33. After that the formulas overflow. Newton needs a decent starting point, and nothing in the method tells you whether you have one.

Brent's method. Keep a bracket, as bisection does, so the root can't escape. At each step, try a fast step, either the secant through the last two points or a parabola through the last three (**inverse quadratic interpolation**). Accept it only if it lands inside the bracket and shrinks it fast enough; otherwise bisect. On the course's call it takes 4 steps; on the strike-200 call it takes 12 and finds 0.200000.

Distance from σ = 20% after each of the first four iterations. Newton starts at 50%; bisection and Brent start from the bracket [1%, 100%].

ITERATION	BISECTION ERROR	NEWTON ERROR	BRENT ERROR
1	3.0e-01	4.2e-03	3.1e-03
2	5.7e-02	2.9e-08	1.9e-04
3	6.6e-02	8.3e-17	1.5e-09
4	4.4e-03	8.3e-17	2.2e-16

In practice: Newton from a sensible start, with a bracket kept on the side; whenever a Newton step would leave the bracket, bisect instead (Exercise 34.7).

Interpolation: forward rates from a few zero rates

Zero rates and forward rates. A zero rate $z(t)$ prices a zero-coupon bond, $P(t) = e^{-z(t)t}$. The **instantaneous forward rate** $f(t)$ is the rate for borrowing over the short interval just after t, agreed today. It is the rate at which the bond price falls: $f(t) = -\frac{d}{dt}\ln P(t) = \frac{d}{dt}\big(z(t)t\big)$, so

$$f(t) = z(t) + t\,z'(t).$$

The forward rate depends on the *slope* of the zero curve, multiplied by t. So small wiggles in an interpolated zero curve become large wiggles in the forwards, and forwards are what decide the price of anything that depends on future rates. **Judge an interpolation by its forwards.**

The data. Made-up zero rates: 3.0%, 3.3%, 3.5%, 3.8% and 4.1% at 1, 2, 3, 5 and 10 years.

Straight lines. Join the points by straight lines. The slope z' jumps at every point, so the forward curve jumps there: just before 2 years it is 3.90%, just after it is 3.70% (Exercise 34.4). The forward curve is a saw.

A natural cubic spline. On each interval use a cubic polynomial, with the value, slope and curvature continuous at every point, and zero curvature at both ends. Writing M_i for the curvature (second derivative) at point i and h_i for the gaps, the continuity of the slope gives one equation per inside point (Exercise 34.3):

$$h_{i-1}M_{i-1} + 2(h_{i-1} + h_i)M_i + h_iM_{i+1} = 6\left(\frac{z_{i+1} - z_i}{h_i} - \frac{z_i - z_{i-1}}{h_{i-1}}\right).$$

Each equation links three neighbours: a tridiagonal system, solved by Lesson 29's method.

The natural cubic spline: build the tridiagonal system and solve it.

```
def natural_spline(x, y):
    h = np.diff(x); n = len(x)
    a = np.zeros(n); b = np.ones(n); c = np.zeros(n); d = np.zeros(n)
    for i in range(1, n - 1):
        a[i], b[i], c[i] = h[i - 1], 2 * (h[i - 1] + h[i]), h[i]
        d[i] = 6 * ((y[i + 1] - y[i]) / h[i] - (y[i] - y[i - 1]) / h[i - 1])
    M = thomas(a, b, c, d)           # second derivatives at the knots; zero at both ends
```

The spline's forwards are smooth, but look at their shape: they rise to 4.433% at 6.0 years, then fall to 4.367% at 9.0 years, even though the zero rates rise steadily. The spline's smoothness has invented a hump.

A monotone cubic. The Fritsch-Carlson method also uses a cubic on each interval, but chooses the slopes at the points with a weighted average that makes the curve rise wherever the data rise: it can't overshoot. Its forwards are continuous, but they still have a dip, from 4.337% at 4.4 years to 4.257% at 5 years.

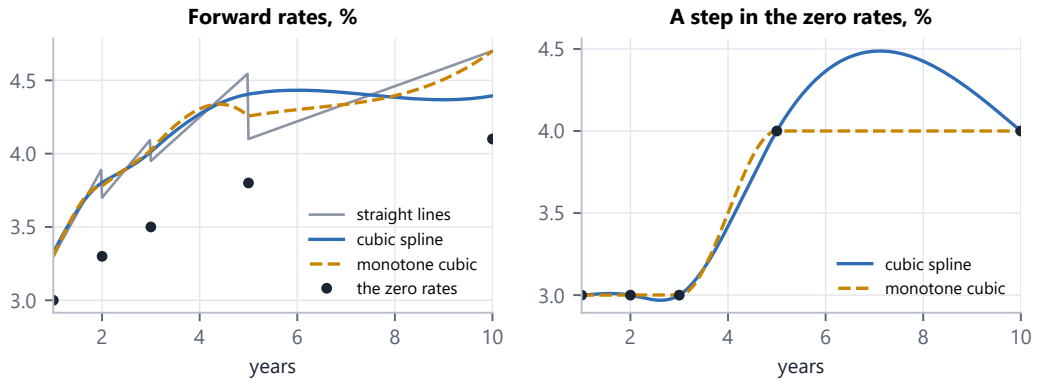


Figure 34.2. Left: forward rates from three ways of interpolating the same five zero rates (black dots, in %). Right: a made-up curve with a step from 3% to 4%; the spline overshoots both levels, the monotone cubic doesn't.

The forward curves on 1 to 10 years.

METHOD	Lowest forward (%)	Highest forward (%)	Forward at 10 years (%)	TURNING POINTS
linear	3.300	4.700	4.700	6
spline	3.324	4.433	4.394	2
mono-tone	3.300	4.700	4.700	2

Overshoot. The right of the picture is a curve with a step: 3% out to 3 years, 4% from 5. The spline swings down to 2.969% and up to 4.487%, outside the data at both ends. The monotone cubic stays between 3.00% and 4.00%.

The lesson. Every interpolation leaves its mark on the forwards: jumps, invented humps, or kinks. There is no free choice. What banks do (Lesson 57) is pick the quantity to interpolate with care, and always look at the forward curve before using it.

Integration: Gaussian quadrature

The idea. A rule $\int_a^b f(x)\,dx \approx \sum_i w_i f(x_i)$ with n points has $2n$ numbers to choose: n positions and n weights. Choose them so that the rule is exact for every polynomial of degree up to $2n-1$ ($2n$ conditions). That is **Gaussian quadrature**. Two points, well placed, integrate every cubic exactly (Exercise 34.5). A smooth integrand is close to a polynomial, so the error falls very fast as n grows.

- **Gauss-Legendre** for $\int_{-1}^{1} f(x)\,dx$, moved to any interval $[a, b]$ by a straight-line change of variable.

- **Gauss-Hermite** for $\int e^{-x^2} f(x)\,dx$ over the whole line, which handles an average against a normal density: $E[g(Z)] \approx \frac{1}{\sqrt{\pi}} \sum_i w_i g(\sqrt{2}\,x_i)$.

The kink. The call's integrand $\max(S_T(z) - K, 0)\varphi(z)$ has a corner at $z^* = -d_- = 0.19395$, where the call starts paying. No polynomial has a corner, so Gauss-Hermite across the whole line converges slowly and unevenly. **Start the interval at the kink**, and the integrand on $[z^*, 8]$ is smooth: Gauss-Legendre then converges at full speed.

Errors in the call price, 7.1281, by each rule.

Points (n)	TRAPEZIUM ON [−8, 8]	GAUSS-HERMITE	GAUSS-LEGENDRE FROM THE KINK
5	-7.1e+00	-4.0e-01	-1.6e-01
10	1.5e+00	1.8e-01	-5.2e-05
20	9.6e-02	1.3e-02	-1.5e-13
40	-9.5e-02	-6.9e-02	-2.6e-13
80	1.3e-02	-5.7e-02	-2.4e-13

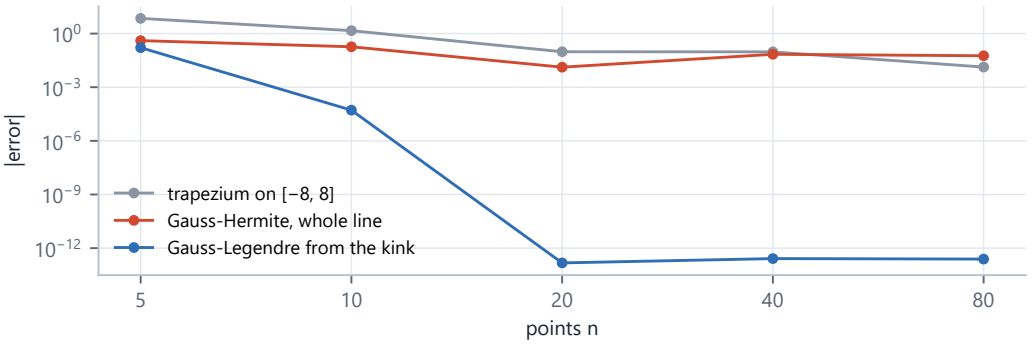


Figure 34.3. The same errors on log scales. With the kink at the edge of the interval, Gauss-Legendre is exact to 13 digits with 20 points; across the kink, neither the trapezium nor Gauss-Hermite gets below about 10^{-2}.

The same idea served Lesson 22's trees: put the kink on a node, or at the edge of an interval, and the method's full accuracy comes back.

Fitting: least squares

Linear least squares. We have data $y_1, \ldots, y_m$ and a model that is a sum of known functions with unknown coefficients: $y \approx X\beta$, where column j of X holds function j at each data point. Choose β to make the sum of squared errors $\|y - X\beta\|^2$ as small as possible. Setting its derivative with respect to β to zero gives the **normal equations** (Exercise 34.6)

$$X^\top X\beta = X^\top y.$$

Geometrically, they say that the leftover errors are at right angles to every column of X: $X^\top(y - X\beta) = 0$. The fit is the closest point to y that the model can reach.

A smile. Made-up implied volatilities at seven strikes, against log-moneyness $k = \ln(K/F)$, with $F = S_0 e^{rT} = 103.0455$. They are made from $0.20 - 0.10k + 0.30k^2$ plus a small fixed noise of about 0.1 volatility point. Fitting the parabola $a + bk + ck^2$ gives $a = 0.2003$, b = -0.1029, $c = 0.2683$, with leftover errors of size 0.0008. The curvature c is the least well pinned down: the strikes span only a small range of k, so k^2 varies little.

Nonlinear least squares. A curve for zero rates with three parameters:

$$z(t) = b_0 + b_1\,\frac{1 - e^{-t/\tau}}{t/\tau}.$$

As $t \to 0$ it tends to $b_0 + b_1$ (the short rate), as $t \to \infty$ to b_0 (the long rate), and τ sets how fast it gets from one to the other. It is linear in b_0 and b_1, but not in τ. The data: made-up zero rates at 1, 2, 3, 5, 7, 10, 20 and 30 years, from $b_0 = 4.2\%$, $b_1 = -1.4\%$, $\tau = 2$, with one basis point of fixed noise.

Gauss-Newton. Linearise the model around the current guess p: $z(p + \delta) \approx z(p) + J\delta$, where the **Jacobian** J holds the derivatives of the model with respect to each parameter. Then δ solves a linear least-squares problem: $J^\top J\,\delta = J^\top(y - z(p))$. Repeat.

Levenberg-Marquardt. Add a damping term:

$$\big(J^\top J + \lambda\,\mathrm{diag}(J^\top J)\big)\,\delta = J^\top\big(y - z(p)\big).$$

With λ near zero it is Gauss-Newton. With λ large the step becomes short and points straight downhill. After each step: if the error fell, accept it and divide λ by 3 (trust the linearisation more); if not, reject it and multiply λ by 3.

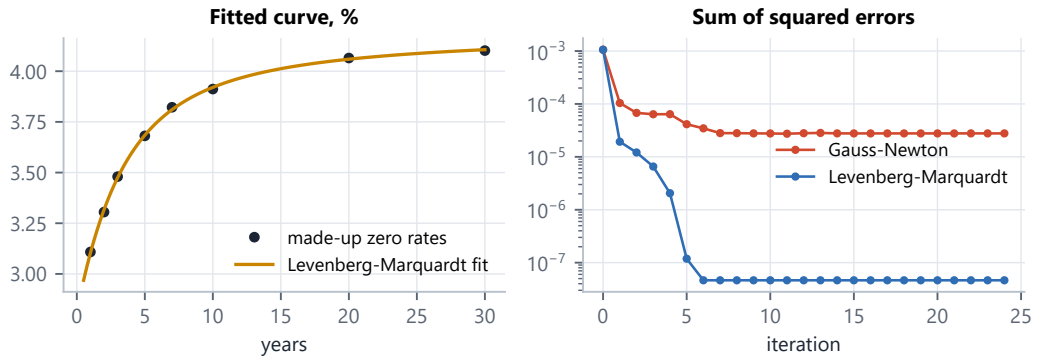


Figure 34.4. Left: the made-up zero rates and the Levenberg-Marquardt fit. Right: the sum of squared errors at each iteration, from the start (5%, −3%, τ = 20). Gauss-Newton stalls three orders of magnitude above Levenberg-Marquardt.

From the same poor start, $(b_0, b_1, \tau) = (5\%, -3\%, 20)$:

- Gauss-Newton runs off. Its full steps drive τ to -2.3e+06 and b_0 to -1356, with b_1 almost exactly $-b_0$: a curve that is nearly flat and fits badly. Its error stalls at 2.8e-05.
- Levenberg-Marquardt shortens the steps that don't help, and reaches $b_0 =$ 0.04200, $b_1 =$ -0.01389, $\tau = 2.028$, with error 4.6e-08: the true values, up to the noise.

What to remember

- **Bisection:** safe, one binary digit per step. **Newton:** $\sigma_{n+1} = \sigma_n - (C - C^*)/\mathcal{V}$, digits double per step, fails when vega is tiny. **Brent:** a bracket plus fast steps; safe and fast.
- **Forwards:** $f(t) = z(t) + t\,z'(t)$. Straight lines make forwards jump; a cubic spline can invent humps and overshoot; a monotone cubic can't overshoot the data, but its forwards still bend. Always look at the forwards.
- **Gaussian quadrature:** exact for polynomials up to degree $2n - 1$; very fast on smooth integrands; slow across a kink, so start an interval there.
- **Least squares:** $X^\top X\beta = X^\top y$. **Nonlinear:** Gauss-Newton linearises; Levenberg-Marquardt damps the step, and accepts only steps that help.

Common mistakes

- **Newton without a fallback.** It can jump far outside any sensible range in one step.
- **Interpolating zero rates and never looking at the forwards.**

- **Applying a high-order rule across a kink.** The order is lost; split the interval at the kink.
- **Fitting prices when the quotes are volatilities**, or the reverse, without thinking about the weights: a cent of error matters far more on a cheap option than on an expensive one.
- **Taking a nonlinear fit's answer without checking where it started**, or whether the parameters still mean something.

Exercises

Solutions are in *Part 5: Solutions*.

34.1 Bisection starts from the bracket $[0.01, 1]$. How many halvings does it need to pin σ down to 10^{-8}? Newton's error roughly squares each step: $e_{n+1} \approx e_n^2$. Starting from an error of 10^{-2}, how many steps does it need to reach 10^{-12}?

34.2 Derive Newton's update $\sigma_{n+1} = \sigma_n - \big(C(\sigma_n) - C^*\big)/\mathcal{V}(\sigma_n)$ from a first-order Taylor expansion. Explain why it fails when vega is nearly zero, as it is for a deep out-of-the-money option at a low starting volatility. What does Brent's method do instead?

34.3 A natural cubic spline through (t_i, y_i) is a cubic on each interval, with matching values, slopes and second derivatives at the knots, and zero second derivative at both ends. Writing M_i for the second derivative at t_i, derive the equation $h_{i-1}M_{i-1} + 2(h_{i-1}+h_i)M_i + h_i M_{i+1} = 6\Big(\frac{y_{i+1}-y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\Big)$, where $h_i = t_{i+1} - t_i$.

34.4 The forward rate is $f(t) = z(t) + t\,z'(t)$. With straight-line interpolation of the zero rates 3.0% (1 year), 3.3% (2 years) and 3.5% (3 years), find the forward rate just before and just after 2 years. Why is a jump in the forward curve a problem for pricing?

34.5 Show that the two-point Gauss-Legendre rule $\int_{-1}^{1} f(x)\,dx \approx f(-1/\sqrt{3}) + f(1/\sqrt{3})$ is exact for 1, x, x^2 and x^3, and not for x^4. Why can a rule with n well-chosen points be exact up to degree $2n-1$?

34.6 Derive the normal equations $X^\top X\beta = X^\top y$ by setting the derivative of $\|y - X\beta\|^2$ to zero. Fit a straight line $y = a + bx$ by hand to the points $(0,1)$, $(1,2)$ and $(2,2)$.

34.7 *(Code)* Build an implied-volatility solver that uses Newton's method and falls back to Brent when a step leaves the bracket. Test it on strikes from 50 to 200 and maturities from 0.1 to 5 years (200 cases) at true volatilities between 10% and 60%. Report the largest error, and how often plain Newton from a 20% start fails.

34.8 *(Code)* Fit the three-parameter curve of this lesson with Levenberg-Marquardt from the starts $(5\%, -3\%, 20)$, $(3\%, 1\%, 0.5)$ and $(4\%, 0\%, 2)$. Which runs fail, and why? Add the bound $\tau \geq 0.05$ and try again.

Self-test

1. Why does Newton's method converge so much faster than bisection, and what does it need that bisection doesn't?
2. Why should you look at forward rates, not zero rates, to judge an interpolation?
3. In Levenberg-Marquardt, what happens to the step as λ grows, and when does λ shrink?

Companion notebook: `s5e34_numerical_methods.ipynb`. Every number in this chapter is computed by it.

LESSON 35 · VIDEO 35

Regression and PDEs, Further

Why a regression estimates a conditional expectation, tested on a put whose answer we know; kernel regression and the bandwidth trade-off; the forward Kolmogorov (Fokker-Planck) equation, which carries the density forward from today, derived and solved on a grid; and ADI schemes for two shares, where the simplest one loses an order of accuracy and one correction restores it.

Why this lesson

Three tools that the later parts of the course depend on:

- **Regression inside a simulation.** Longstaff-Schwartz (Lesson 30) estimated a continuation value by regressing future cash flows on today's share price. Why does that work, and how good is the estimate? Lesson 56 (local-stochastic volatility) needs a more flexible regression still: a **kernel** regression.
- **The forward equation.** Feynman-Kac (Lesson 28) gives an equation that runs *backwards* from a payoff. The density of the share price obeys an equation that runs *forwards* from today. Calibrating volatility surfaces (Lessons 51 and 56) needs this one.
- **Two dimensions.** Options on two shares, and two-factor interest-rate models (Lessons 66 and 67), give PDEs in two space variables. The implicit schemes of Lesson 29 become expensive there. **ADI** schemes keep the cost down.

What you need from earlier lessons

- **Conditional expectation and the tower property** (Lesson 8). $E\big[E[Y \mid X]\, g(X)\big] = E[Y\, g(X)]$ for any function g.
- **Longstaff-Schwartz** (Lesson 30).
- **Feynman-Kac** (Lesson 28). $g(t,x) = E[h(X_T) \mid X_t = x]$ solves $g_t + a\, g_x + \frac{1}{2} b^2 g_{xx} = 0$.
- **Finite differences** (Lesson 29). Crank-Nicolson, the tridiagonal (Thomas) solver, and the explicit stability limit.
- **Margrabe's formula** (Lesson 27), and correlated draws by Cholesky (Lesson 31).

The picture

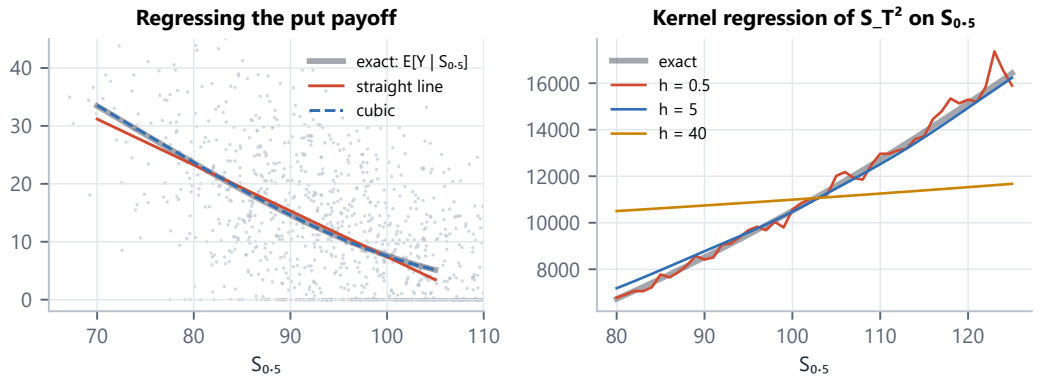


Figure 35.1. Left: 1,500 simulated pairs ($S_{0.5}$, discounted put payoff), the exact conditional expectation (the Black-Scholes put with half a year left, grey), and two regressions on the in-the-money paths. Right: kernel regressions of S_T^2 on $S_{0.5}$ with three bandwidths, against the exact curve.

A cloud of noisy points, and a smooth curve through its middle. That curve, the average of Y for each value of X, is the conditional expectation, and every method in the first half of this lesson is a way of estimating it.

Regression as conditional expectation

Claim. Among all functions g, the one that makes $E\big[(Y - g(X))^2\big]$ smallest is $g(X) = E[Y \mid X]$.

Why (Exercise 35.1 writes it out). Write $Y - g(X) = \big(Y - E[Y \mid X]\big) + \big(E[Y \mid X] - g(X)\big)$ and square. The cross term has mean zero by the tower property, since the second bracket is a function of X. So

$$E\big[(Y - g(X))^2\big] = E\big[(Y - E[Y \mid X])^2\big] + E\big[(E[Y \mid X] - g(X))^2\big],$$

and only the second term depends on g. It is smallest, zero, when $g = E[Y \mid X]$. ■

What a regression does. A least-squares regression on a few basis functions (Lesson 34) finds the g that makes the *sample* average of $(Y - g(X))^2$ smallest, among the functions it can build. So it estimates $E[Y \mid X]$, as well as that family allows, from the data available.

A test where we know the answer. Simulate 20,000 paths of the share to $t = 0.5$ and on to $T = 1$. Let Y be the discounted put payoff, $e^{-r(T-t)}\max(K - S_T, 0)$. Then $E[Y \mid S_{0.5}]$ is the Black-Scholes put with half a year left. Each Y is very noisy: its standard de-

viation around that curve is 8.27. At $t = 0.5$, 0.62 of the paths are in the money. Regress Y on polynomials in $S_{0.5}$, either on all paths or only on the in-the-money ones, and measure the error against the exact curve where it matters, on the in-the-money paths:

Root-mean-square error of the fitted continuation value against the exact put, over the in-the-money paths.

DEGREE	FITTED ON ALL PATHS	FITTED ON IN-THE-MONEY PATHS
1	2.6768	0.7515
2	0.3057	0.1734
3	0.3884	0.0626
4	0.0983	0.1248
5	0.1084	0.1415

Reading the table.

- A straight line is far too crude, especially when fitted to all paths: it must bend to follow the flat region above the strike, where the put is nearly worthless, and that ruins it in the money.
- With the cubic, fitting only where it matters does best: an error of 0.0626, against each payoff's noise of 8.27.
- More terms don't help. With degrees 4 and 5, the fit on the in-the-money paths gets *worse*: the extra flexibility starts to follow the noise. That is why Longstaff-Schwartz uses a few terms, fitted only where exercise is possible (Exercise 35.2).

Kernel regression

A polynomial fixes the shape of the answer in advance. A **kernel regression** lets the data decide. To estimate $E[Y \mid X = x]$, take a weighted average of the Y_i, with weights that fall off with the distance between X_i and x:

$$\hat{m}(x) = \frac{\sum_i K_h(x - X_i)\, Y_i}{\sum_i K_h(x - X_i)}, \qquad K_h(d) = e^{-d^2/(2h^2)}.$$

This is the **Nadaraya-Watson** estimator. The **bandwidth** h sets how far "nearby" reaches.

The trade-off. With a small h, only a few points near x count: the estimate follows the true curve closely but is noisy (high **variance**). With a large h, many points count, so the noise averages out, but they come from a wide range of x where the true curve is different: the estimate is flattened (high **bias**). As $h \to 0$ the estimate at a data point becomes that point's own Y; as $h \to \infty$ it becomes the overall average (Exercise 35.3).

A test. $E[S_T^2 \mid S_{0.5}] = S_{0.5}^2 e^{(2r+\sigma^2)(T-0.5)}$ (Lesson 17): $S_{0.5}^2$ times 1.0513. With 5,000 paths:

Kernel regression of S_T² on $S_{0.5}$, on $S_{0.5}$ from 80 to 125. The values being estimated are around 10,000.

Bandwidth (h)	ROOT-MEAN-SQUARE ERROR	AVERAGE ERROR AT THE TWO ENDS
0.5	324	-238
2.0	127	198
5.0	233	131
15.0	1459	-282
40.0	2526	-489

The error is smallest at $h = 2$. Below that the estimate wobbles; above, it flattens, and at $h = 40$ it is nearly a straight line. Near the ends of the data the bias is worst, because all the neighbours lie on one side.

Cost. Each estimate looks at every path, so estimating at m points from n paths takes mn operations. In practice, estimate on a modest grid of x values and interpolate between them (Exercise 35.7).

The forward Kolmogorov equation

Backward and forward. Lesson 28's equation for $g(t,x) = E[h(X_T) \mid X_t = x]$ runs **backwards**: it starts from the payoff at T and gives the value for every starting point. The **density** $p(t,x)$ of X_t, given today's value, carries the same information the other way round. It obeys an equation that runs **forwards** from today.

Claim (the **forward Kolmogorov**, or **Fokker-Planck**, equation). For $dX = a(X)\,dt + b(X)\,dW$,

$$\boxed{p_t = -\big(a\,p\big)_x + \tfrac{1}{2}\big(b^2 p\big)_{xx}}$$

Derivation.

Step 1. A test function. Take any smooth function $\phi(x)$ that vanishes far out. By Itô's lemma, $d\phi(X_t) = \big(a\phi' + \tfrac{1}{2}b^2\phi''\big)\,dt + b\phi'\,dW$. Take expectations; the dW term has mean zero (Lesson 16):

$$\frac{d}{dt}E[\phi(X_t)] = E\big[a\phi' + \tfrac{1}{2}b^2\phi''\big].$$

Step 2. Write both sides with the density. The left side is $\int \phi(x)\,p_t(t,x)\,dx$. The right side is $\int \big(a\phi' + \tfrac{1}{2}b^2\phi''\big)p\,dx$.

Step 3. Move the derivatives onto p by integrating by parts (once for ϕ', twice for ϕ''). The boundary terms vanish because ϕ and p vanish far out:

$$\int \phi\, p_t\, dx = \int \phi\Big(-(ap)_x + \tfrac{1}{2}(b^2 p)_{xx} \Big) dx.$$

Step 4. This holds for every test function ϕ, so the two brackets must be equal. ■

For constant a and b the equation is $p_t = -a\,p_x + \frac{1}{2}b^2 p_{xx}$, and a normal density with mean $x_0 + at$ and variance $b^2 t$ solves it (Exercise 35.4).

Solving it on a grid. For $X = \ln S$ under the risk-neutral probability, $a = \nu = r - \frac{1}{2}\sigma^2$ and $b = \sigma$. Today the density is a spike at $\ln S_0$, which no grid can hold, so start a moment later, at $t_0 = 0.02$, from the exact (narrow) normal density. Then step forward to $T = 1$ with Crank-Nicolson (Lesson 29) on 600 space steps and 400 time steps, each step a tridiagonal solve.

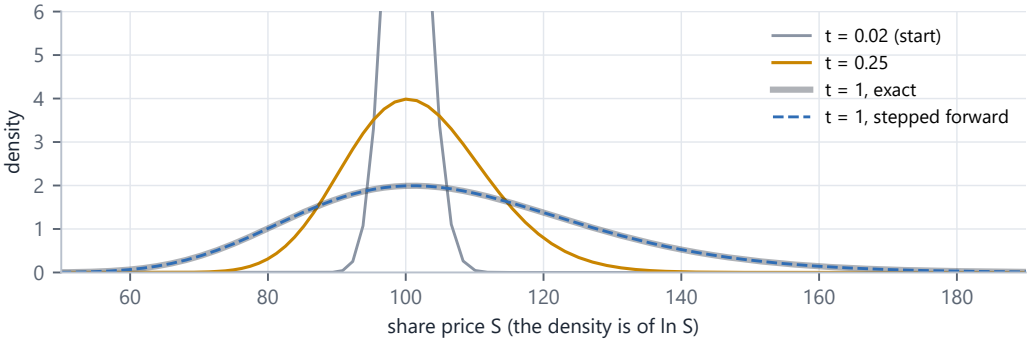


Figure 35.2. The density of ln S_t, plotted against S, as the forward equation carries it from t = 0.02 to 1. At t = 1 it lies on the exact normal density.

Checks at $T = 1$**.**

- The total probability is 1.00000000: the equation conserves it (Exercise 35.5).
- The average share price, $\int e^x p\, dx$, is 103.0455, against the forward $S_0 e^{rT} =$ 103.0455.
- The largest error in the density is 9.6e-05, against a peak of 1.995.

A price from the density. $e^{-rT}\int \max(e^x - K, 0)\, p(T,x)\, dx =$ 7.1279, against the formula's 7.1281, a difference of -1.1e-04.

Why have both. The backward equation gives *one payoff* for *every starting point*. The forward equation gives *every payoff* for *one starting point*, today's. When calibrating a model to many quoted options at once (Lessons 51 and 56), the forward one does all the work in one solve.

Two dimensions: the ADI scheme

The equation. For an option on two shares, with $x = \ln S_1$ and $y = \ln S_2$ (Exercise 35.6),

$$U_\tau = \nu_1 U_x + \tfrac{1}{2}\sigma_1^2 U_{xx} + \nu_2 U_y + \tfrac{1}{2}\sigma_2^2 U_{yy} + \rho\sigma_1\sigma_2 U_{xy} - rU, \qquad \nu_i = r - \tfrac{1}{2}\sigma_i^2.$$

The new term is the **cross derivative** U_{xy}, which carries the correlation. On a grid with M steps in each direction, it is approximated from the four diagonal neighbours: $U_{xy} \approx \big(U_{i+1,j+1} - U_{i+1,j-1} - U_{i-1,j+1} + U_{i-1,j-1}\big)/(4h_x h_y)$.

Why two dimensions is harder. An explicit step is cheap, but stable only for $\Delta\tau \lesssim h^2/(\sigma_1^2 + \sigma_2^2)$: halving h needs four times the steps, as in Lesson 29. A fully implicit step is stable, but the M^2 unknowns now form a system that is not tridiagonal: each point is linked to neighbours in both directions. Solved directly, it costs about M^4 operations per step.

ADI: implicit in one direction at a time. Split the right-hand side into three parts: A_1 (the x terms and half of $-rU$), A_2 (the y terms and the other half) and A_0 (the cross term). The **Douglas** scheme, with $\theta = \frac{1}{2}$, takes each time step in three stages:

$$\begin{aligned} Y_0 &= U + \Delta\tau\,(A_0 + A_1 + A_2)\,U && \text{(an explicit step with everything)} \\ (I - \theta\Delta\tau A_1)\,Y_1 &= Y_0 - \theta\Delta\tau A_1 U && \text{(correct implicitly in } x\text{)} \\ (I - \theta\Delta\tau A_2)\,Y_2 &= Y_1 - \theta\Delta\tau A_2 U && \text{(correct implicitly in } y\text{)} \end{aligned}$$

and $U_{\text{new}} = Y_2$. Each correction links points along one direction only, so it is a batch of M independent tridiagonal systems: about M^2 operations for the whole step, like an explicit step, but with the stability of an implicit one. The cross term stays explicit.

The catch, and the repair. With a cross term, Douglas is only **first order** in time. **Craig-Sneyd** adds one more stage: recompute the cross term with the new values, and repeat the two corrections:

$$\tilde{Y}_0 = Y_0 + \tfrac{1}{2}\Delta\tau\big(A_0 Y_2 - A_0 U\big), \quad \text{then the same two corrections, starting from } \tilde{Y}_0.$$

That restores **second order**, for twice the work per step.

One time step of Douglas, and the extra Craig-Sneyd stage.

```
    for step in range(N_):
        th = 1.0 if step < damp else 0.5
        a0U, a1U, a2U = A0(U), A1(U), A2(U)
        Y0 = U + k * (a0U + a1U + a2U)
        U_new = sweeps(Y0, a1U, a2U, th)
        if scheme == "craig-sneyd":                                  # refresh the cross term
and sweep again
            U_new = sweeps(Y0 + 0.5 * k * (A0(U_new) - a0U), a1U, a2U, th)
        U = U_new
```

The test: Margrabe's exchange option. Lesson 27's option to swap share 2 for share 1, $\max(S_1 - S_2, 0)$, with $S_1 = 100$, $S_2 = 95$, $\sigma_1 = 20\%$, $\sigma_2 = 30\%$ and $\rho = 0.5$, is worth 12.9523 by Margrabe's formula. Two lessons from earlier chapters matter here again:

- **Put the kink on the grid** (Lessons 22 and 29). The payoff has a kink along the line $S_1 = S_2$. Use one spacing h in both directions, with $h = \ln(100/95)/m$ for a whole number m: then the line $x = y$ passes exactly through grid points.
- **Damp the kink** (Lesson 29). Crank-Nicolson's $\theta = \frac{1}{2}$ damps a kink's zigzag only slowly, so take the first four steps with $\theta = 1$.

Errors against Margrabe's 12.9523, with as many time steps as grid steps in each direction, on ln S ± 1.5.

Divisor (m)	Grid steps (M)	Time steps (N)	DOUGLAS ERROR	CRAIG-SNEYD ERROR	STEPS AN EXPLICIT SCHEME WOULD NEED
1	60	60	-2.5e-02	-8.7e-03	50
2	118	118	-1.1e-02	-2.1e-03	198
4	234	234	-5.0e-03	-5.1e-04	791

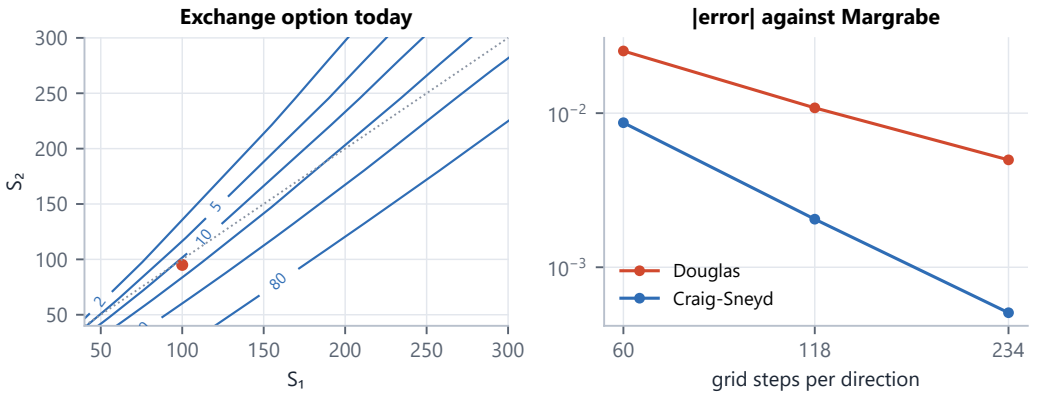


Figure 35.3. Left: the exchange option's value today over a range of (S_1, S_2); the red dot is today's prices, the dotted line $S_1 = S_2$. Right: the error of each scheme as the grid is refined, on log scales.

Reading the table. Each time h halves (with the time step halving too), Douglas's error falls by 2.17 and Craig-Sneyd's by 4.05: first order against second. On the finest grid, Craig-Sneyd is ten times more accurate for twice the work per step. An explicit scheme on that grid would need 791 time steps just to be stable.

What to remember

- **Regression:** $E[Y \mid X]$ is the best predictor of Y from X in mean square; least squares on basis functions estimates it. Fit where it matters (in the money for Longstaff-Schwartz), and with few terms.
- **Kernel regression:** a locally weighted average; a small bandwidth gives noise, a large one gives bias.
- **Forward equation:** $p_t = -(ap)_x + \frac{1}{2}(b^2 p)_{xx}$, from today's density forwards; the backward equation runs from the payoff. Forward gives all payoffs from one start; backward gives one payoff from all starts.
- **ADI:** an explicit step, then implicit corrections one direction at a time, each a batch of tridiagonal solves. Douglas is first order with a cross term; Craig-Sneyd's extra stage makes it second order.
- **Kinks:** put them on the grid, and damp them with a few fully implicit steps.

Common mistakes

- **Regressing on all paths in Longstaff-Schwartz.** The fit bends to follow paths where exercise is impossible.
- **Adding more basis functions to get a better fit.** Past a point, they fit the noise.
- **Choosing the bandwidth by eye.** Measure the error, or cross-validate.
- **Mixing up the two equations.** The forward one has derivatives of $a\,p$ and $b^2 p$, and minus signs, and runs from today.
- **Dropping or mis-scaling the cross term.** Without $\rho\sigma_1\sigma_2 U_{xy}$ the shares are priced as independent.

Exercises

Solutions are in *Part 5: Solutions*.

35.1 Show that $E\left[(Y - g(X))^2\right]$ is smallest, over all functions g, when $g(X) = E[Y \mid X]$. (Write $Y - g = (Y - E[Y \mid X]) + (E[Y \mid X] - g)$, expand, and use the tower property on the cross term.)

35.2 In Longstaff-Schwartz (Lesson 30) the regression uses only the paths that are in the money at that date. Explain why. What goes wrong with the fitted continuation value, where it matters, if the out-of-the-money paths are included?

35.3 Three paths have $(X, Y) = (90, 10)$, $(100, 0)$ and $(110, 2)$. Compute the Nadaraya-Watson estimate at $x = 100$ with a normal kernel of bandwidth $h = 10$, and its limits as $h \to 0$ and $h \to \infty$. What does that say about choosing h?

35.4 Show that the normal density with mean $x_0 + at$ and variance $b^2 t$ solves the forward equation $p_t = -a\,p_x + \tfrac{1}{2} b^2 p_{xx}$.

35.5 Using the forward equation and integration by parts, show that $\int p(t,x)\,dx$ doesn't change with t. Then show that if g solves the backward equation $g_t + a\,g_x + \tfrac{1}{2} b^2 g_{xx} = 0$, the quantity $\int g(t,x)\,p(t,x)\,dx$ doesn't change either. Why does that make forward and backward pricing agree?

35.6 Change variables to $x = \ln S_1$, $y = \ln S_2$ in the two-asset equation $V_t + \tfrac{1}{2}\sigma_1^2 S_1^2 V_{11} + \tfrac{1}{2}\sigma_2^2 S_2^2 V_{22} + \rho\sigma_1\sigma_2 S_1 S_2 V_{12} + r S_1 V_1 + r S_2 V_2 = rV$, and derive the equation of this lesson. Then compare the work per time step of a full implicit solve (a banded system of M^2 unknowns with bandwidth M) and of ADI.

35.7 *(Code)* In Longstaff-Schwartz for the American put, replace the polynomial regression with a kernel regression, and compare the prices with Lesson 30's at 50 exercise dates. How sensitive is the answer to the bandwidth?

35.8 *(Code)* Use the Craig-Sneyd scheme for a call on the average of the two shares, $\max\left(\tfrac{1}{2}(S_1 + S_2) - 100, 0\right)$, with the lesson's parameters, and check the price by Monte Carlo with correlated draws (Lesson 31).

Self-test

1. A regression of discounted payoffs on today's share price estimates what quantity?
2. Which way in time does the forward Kolmogorov equation run, and what does it carry?
3. Why does an ADI step cost about as much as an explicit one, and what does Craig-Sneyd add?

Companion notebook: `s5e35_regression_and_pdes.ipynb`. Every number in this chapter is computed by it.

Formula sheet and glossary

Formula sheet

Linear algebra (Lesson 31)

- Portfolio variance: $\mathrm{Var}(w^\top R) = w^\top \Sigma w$, with $\Sigma_{ij} = \mathrm{Cov}(R_i, R_j)$.
- Valid covariance or correlation matrix: positive semi-definite, $w^\top \Sigma w \geq 0$ for all w; equivalently, no negative eigenvalue.
- Cholesky: $\Sigma = LL^\top$, L lower triangular; $X = LZ$ has covariance Σ. $L_{jj} = \sqrt{\Sigma_{jj} - \sum_{k<j} L_{jk}^2}$, $L_{ij} = (\Sigma_{ij} - \sum_{k<j} L_{ik}L_{jk})/L_{jj}$.
- Two shares: $X_1 = \sigma_1 Z_1$, $X_2 = \sigma_2(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2)$.
- Eigen-decomposition: $\Sigma = Q\Lambda Q^\top$, $Q^\top Q = I$; equal correlation ρ: eigenvalues $1 + (n-1)\rho$ and $1 - \rho$.
- Equal-weighted basket: $\sigma_{\text{basket}} = \sigma\sqrt{(1 + (n-1)\rho)/n} \to \sigma\sqrt{\rho}$.

Ordinary differential equations (Lesson 32)

- $y' + ay = b(t)$: $y(t) = y(0)e^{-at} + \int_0^t e^{-a(t-u)} b(u)\, du$; constant b: $y \to b/a$.
- Vasicek: $E[r_t] = \bar{r} + (r_0 - \bar{r})e^{-\kappa t}$; $\mathrm{Var}(r_t) = \frac{s^2}{2\kappa}(1 - e^{-2\kappa t})$; $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$.
- Riccati $y' = a + by + cy^2$; $y = -u'/(cu)$ gives $u'' - bu' + acu = 0$.
- CIR bond: $B' = 1 - \kappa B - \frac{1}{2}s_c^2 B^2$, $A' = -\kappa\bar{r}B$; $\gamma = \sqrt{\kappa^2 + 2s_c^2}$, $B(\tau) = \frac{2(e^{\gamma\tau}-1)}{(\gamma+\kappa)(e^{\gamma\tau}-1)+2\gamma}$.
- Euler: error $\propto h$. Runge-Kutta 4: $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, error $\propto h^4$.

Complex numbers and Fourier (Lesson 33)

- $i^2 = -1$; $z = a + bi = |z|e^{i\theta}$; $1/z = \bar{z}/|z|^2$; $e^{i\theta} = \cos\theta + i\sin\theta$.
- Characteristic function $\varphi(u) = E[e^{iuX}]$; $N(m, s^2)$: $e^{ium - \frac{1}{2}s^2u^2}$; $\ln S_T$: $m = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$, $s^2 = \sigma^2 T$.
- Density: $f(x) = \frac{1}{\pi}\int_0^\infty \mathrm{Re}[e^{-iux}\varphi(u)]\, du$.
- Gil-Pelaez: $P(X > k) = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[e^{-iuk}\varphi(u)/(iu)\right] du$.
- Share measure: $\varphi_S(u) = \varphi(u - i)/\varphi(-i)$; $C = S_0 P^S(S_T > K) - Ke^{-rT}\tilde{P}(S_T > K)$.

- A step du can't distinguish x from $x + 2\pi/du$ (aliasing).

Numerical methods (Lesson 34)

- Newton: $\sigma_{n+1} = \sigma_n - (C(\sigma_n) - C^*)/\mathcal{V}(\sigma_n)$; error roughly squares each step.
- Bisection: width $(b-a)/2^n$ after n steps.
- Forward rate: $f(t) = z(t) + t\,z'(t)$.
- Natural cubic spline: $h_{i-1}M_{i-1} + 2(h_{i-1}+h_i)M_i + h_i M_{i+1} = 6\left(\frac{y_{i+1}-y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right)$, $M_0 = M_n = 0$.
- Gaussian quadrature: n points, exact up to degree $2n-1$; Gauss-Hermite: $E[g(Z)] \approx \frac{1}{\sqrt{\pi}} \sum_i w_i g(\sqrt{2}x_i)$.
- Least squares: $X^\top X\beta = X^\top y$. Gauss-Newton: $J^\top J\delta = J^\top(y - z)$. Levenberg-Marquardt: $(J^\top J + \lambda\,\mathrm{diag}(J^\top J))\delta = J^\top(y-z)$.

Regression and PDEs (Lesson 35)

- $E[Y \mid X]$ minimises $E[(Y - g(X))^2]$.
- Nadaraya-Watson: $\hat{m}(x) = \sum_i K_h(x - X_i)Y_i / \sum_i K_h(x - X_i)$, $K_h(d) = e^{-d^2/(2h^2)}$.
- Forward (Fokker-Planck): $p_t = -(ap)_x + \frac{1}{2}(b^2 p)_{xx}$. Backward: $g_t + a g_x + \frac{1}{2}b^2 g_{xx} = 0$.
- Two shares in logs: $U_\tau = \nu_1 U_x + \frac{1}{2}\sigma_1^2 U_{xx} + \nu_2 U_y + \frac{1}{2}\sigma_2^2 U_{yy} + \rho\sigma_1\sigma_2 U_{xy} - rU$.
- Douglas: $Y_0 = U + \Delta\tau A U$; $(I - \theta\Delta\tau A_j)Y_j = Y_{j-1} - \theta\Delta\tau A_j U$ for $j = 1, 2$. Craig-Sneyd: $\tilde{Y}_0 = Y_0 + \frac{1}{2}\Delta\tau(A_0 Y_2 - A_0 U)$, then the same two corrections.

The course's numbers

Share price today, S_0	100
Strike, K	105
Risk-free rate, r (continuous)	3%
Volatility, σ	20%
Horizon, T	1 year
Call	7.1281
Second share (Lesson 27)	$S_0 = 95$, $\sigma = 30\%$, correlation 0.5
Vasicek (Lesson 28)	$\kappa = 0.5$, $\bar{r} = 4\%$, $s = 1\%$, $r_0 = 3\%$
Exchange option (Margrabe)	12.9523

Glossary

ADI (alternating direction implicit). A scheme for PDEs in several dimensions that takes an explicit step, then corrects it implicitly in one direction at a time.

Aliasing. A sum with step du can't distinguish x from $x + 2\pi/du$; parts of a distribution that far away are folded back on top.

Bandwidth. The width h of a kernel: how far "nearby" reaches.

Bisection. Root finding by repeatedly halving a bracket in which the function changes sign.

Brent's method. Root finding that keeps a bracket but tries faster interpolation steps.

Characteristic function. $\varphi(u) = E[e^{iuX}]$; it always exists and determines the distribution.

Cholesky factor. A lower-triangular L with $LL^\top = \Sigma$.

Complex number. $a + bi$, with $i^2 = -1$.

Covariance matrix. The table of all covariances $\mathrm{Cov}(R_i, R_j)$.

Craig-Sneyd scheme. Douglas plus one more stage that updates the cross term; second order.

Cubic spline. Piecewise cubic interpolation with continuous value, slope and curvature.

Douglas scheme. The simplest ADI scheme; first order in time when there is a cross derivative.

Eigenvalue, eigenvector. $Av = \lambda v$: a direction the matrix only stretches, and by how much.

Euler's formula. $e^{i\theta} = \cos\theta + i\sin\theta$.

Forward Kolmogorov (Fokker-Planck) equation. The equation that carries the density forward from today.

Forward rate. The rate for borrowing over a short interval in the future, agreed today: $f = z + tz'$.

Gaussian quadrature. An integration rule whose points and weights are chosen to be exact for polynomials of the highest possible degree.

Gil-Pelaez formula. $P(X > k)$ from the characteristic function.

Integrating factor. The multiplier e^{at} that turns $y' + ay$ into a single derivative.

Jacobian. The matrix of derivatives of a model's outputs with respect to its parameters.

Kernel regression (Nadaraya-Watson). A locally weighted average of the Y's, estimating $E[Y \mid X = x]$.

Least squares. Choosing parameters to minimise the sum of squared errors.

Levenberg-Marquardt. Nonlinear least squares with a damped Gauss-Newton step.

Modulus, argument. The length and the angle of a complex number.

Monotone cubic. Cubic interpolation whose slopes are chosen so that it can't overshoot the data.

Newton's method. Root finding by following the tangent line.

Normal equations. $X^\top X\beta = X^\top y$.

Positive (semi-)definite. $w^\top A w > 0$ (≥ 0) for every non-zero w.

Principal component analysis (PCA). Describing risks along the eigenvectors of their covariance matrix, which are uncorrelated.

Riccati equation. $y' = a + by + cy^2$; it can blow up in finite time.

Runge-Kutta 4. A fourth-order ODE method averaging four slopes per step.

Step doubling. Estimating a step's error by redoing it as two half steps.

Unwrapping. Following the angle of a complex number continuously, adding 2π where the principal value would jump.

Further reading

- Gilbert Strang, *Introduction to Linear Algebra*: matrices, positive definiteness, Cholesky and eigenvalues, in full.
- William H. Press, Saul A. Teukolsky, William T. Vetterling and Brian P. Flannery, *Numerical Recipes*: ODE solvers, root finding (including Brent), interpolation, quadrature and Levenberg-Marquardt.
- Jim Gatheral, *The Volatility Surface*: Fourier pricing and the Heston model, where Lesson 33's tools are used.
- Daniel J. Duffy, *Finite Difference Methods in Financial Engineering*: ADI schemes, including Douglas and Craig-Sneyd.

- Karel in 't Hout and Sören Foulon, "ADI Finite Difference Schemes for Option Pricing in the Heston Model with Correlation" (2010): a careful comparison of ADI schemes with a cross term.
- Fred N. Fritsch and Ralph E. Carlson, "Monotone Piecewise Cubic Interpolation" (1980).